Bovine Infectious Disease

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Gastrointestinal Diseases

Bovine Salmonellosis

Bovine Salmonellosis

- Numerous serotypes of *Salmonella* spp. have been isolated from cattle, and many of them are considered infectious in cattle.
- Most important: *S* Typhimurium & ***S* Dublin *(both are zoonotic)***
- *S* Typhimurium can cause diseases in a variety of animals, including humans, cattle, sheep, and pigs
- ***S* Dublin is cattle-adapted and has a greater tendency to cause invasive/systemic disease.**.
- Usually **endemic**, sometime causes major outbreak.
- **Multidrug-resistance** are common in US & Canada isolates

法國-沙門氏菌

資料來源：ProMED ，2020/2/12

法國傳出疫情，迄今累計7個地區13例Salmonella Dublin，其中9例需住院，疑似因食用Jean Perrin公司所製造受汙染奶酪所致，當局呼籲勿食用批號FR 25-155-001 CE之產品；經調查，其相關產品已供應至超市及傳統市場，該公司已停售並召回商品。

最後更新日期 2020/2/14

Table 1. *Examples of* Salmonella *subspecies I serotypes adapted to higher vertebrates*

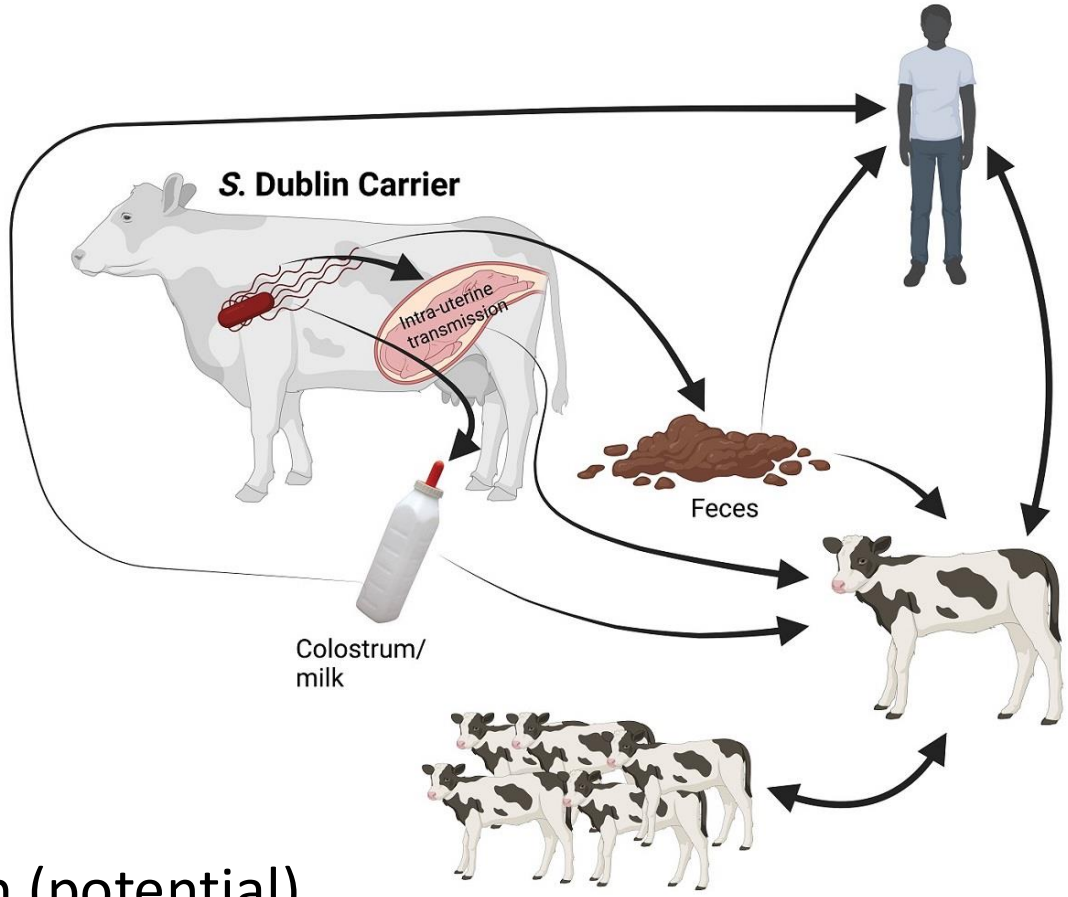
Serotype	Natural host	Other host(s) rarely infected
Host adapted serotype		
Choleraesuis	Swine	Human
Dublin	Bovine	Human, ovine
Host restricted serotype		
Typhi	Human	—
Paratyphi A	Human	—
Paratyphi C	Human	—
Sendai	Human	—
Abortusovis	Ovine	—
Gallinarum	Poultry	—
Typhisuis	Swine	—
Abortusequi	Equine	—

"Unrestricted serotypes, including Typhimurium and Enteritidis, tend to cause enteritis in young animals in a wide range of species. Host-adapted serotypes, such as Dublin and Cholerasuis, can cause either enteritis or systemic disease in their specific species."

Uzzau et al, 2000

Transmission

- Fecal-oral route
  - -- Active carrier
  - -- Symptomless carrier
  - -- **Latent carrier**
- Vertical route (*in utero*)
- Aerosolized transmission (potential)



Velasquez et al., 2023

Pathogenesis & Immunity

- GI tract → mesenteric LN → systemic dissemination
- ***S* Dublin** has adapted to evade host immune response
- Virulence factors: *Salmonella* Pathogenicity Islands (SPI), toxins, flagella (motility), fimbriae (adhesion)…
- ***S* Dublin** encodes the Type III Secretion System from SPI-1 and SPI-2: invade the intestine and spread to systemic sites, respectively
- **Host immunity** is crucial in the outcome of clinical salmonellosis
  -- Young calves with an immature immune system → high morbidity and high mortality

Vaccination and colostrum are important!

Clinical Presentation

- More severe in *S* Dublin comparing to other Salmonella serovar
- **Vary,** depending on age, herd condition…
- All ages are susceptible but most common in calves at 2–12 weeks of age
- In naive herds, the pathogen is rapidly transmitted, and an outbreak ensues.
- Although most Salmonella infections present as GI disease (so diarrhea is common), *S* Dublin infection most often presents as **respiratory illness**.

Clinical Presentation

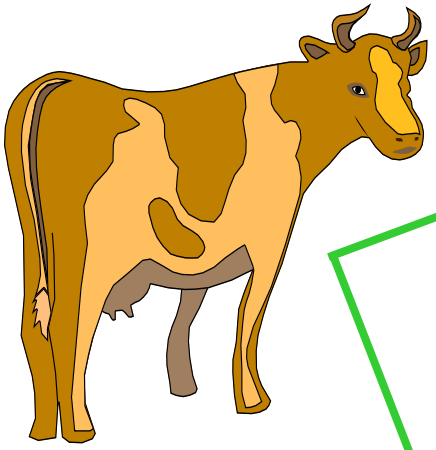
In calves:

- Hyperthermia
- Obtundation (listlessness)
- Anorexia
- **Pneumonia**
- Respiratory distress
- Dehydration
- Septicemia
- Sudden death (peracute)
- Development to chronic infection (retardation, arthritis, loose stool…)

In adult cattle:

- Slight fever
- Mild diarrhea
- Sudden decrease in milk production
- Rarely, bloody diarrhea
- Generation of **latent carriers**

Outcome of a *Salmonella* infection ….



Fever, diarrhea, bloody stool, dehydration, anorexia/ emaciation rapid breathing, sloughing of skin from extremities

No clinical signs

recover

Clinical symptoms

death

No Salmonella in feces and milk

Salmonella infrequently present in feces and or milk

Symptomless Carrier

Salmonella present in the body but not excreted

Latent Carrier

Salmonella in feces and or milk

Active Carrier

Slide provided by 鄭益謙老師





(From Blowey and Weaver, 1991)

Bloody, mucus

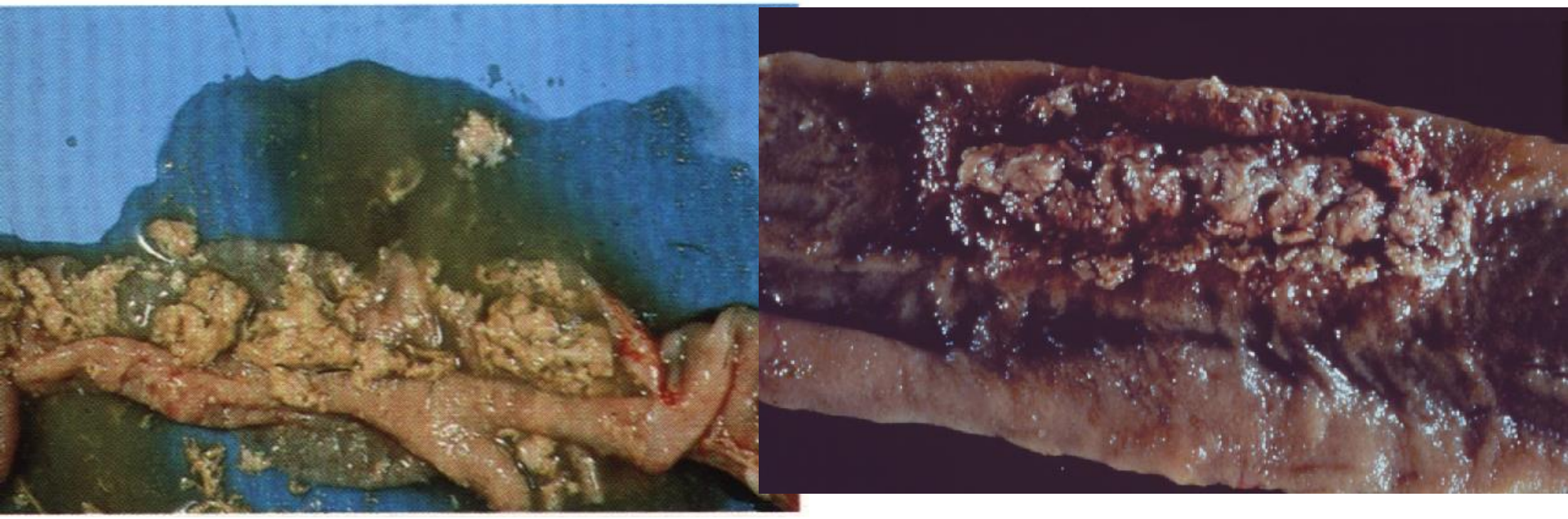
沙氏桿菌性急性腸炎



(From:馮翰鵬, 2002)

Slide provided by 鄭益謙老師

沙氏桿菌性急性腸炎



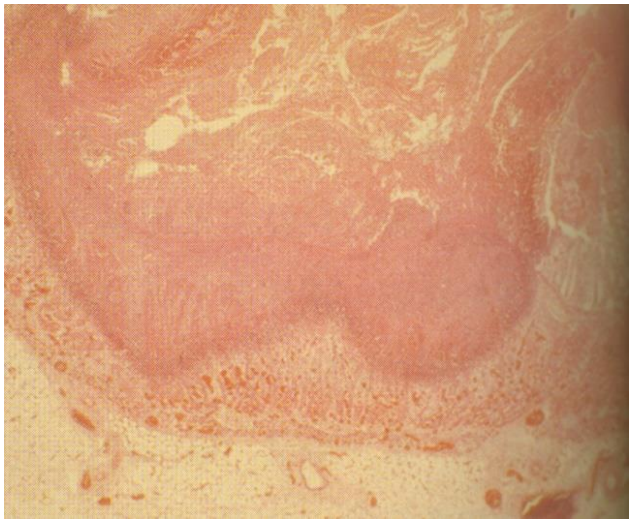
(From Blowey and Weaver, 1991)

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沙氏桿菌性急性腸炎







(From:**馮翰鵬**, 2002)

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牛沙門氏桿菌症

在急性腸炎後期(尤其是S Dublin)常在體端如四肢、耳尖、尾尖發生乾性壞疽(terminal dry gangrene)。





possible immune-mediated reaction

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Brief Communication

Journal of Veterinary Diagnostic Investigation
2017, Vol. 29(6) 860–864
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DOI: 10.1177/1040638717712757
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Histopathology case definition of naturally acquired *Salmonella enterica* serovar Dublin infection in young Holstein cattle in the northeastern United States

Heidi L. Pecoraro, Belinda Thompson, Gerald E. Duhamel[1]

"We propose a histopathology case definition of Salmonella Dublin in <6-mo-old Holstein cattle that includes a combination of **pulmonary alveolar capillary neutrophilia** with or without hepatocellular necrosis and paratyphoid granulomas, splenitis, and lymphadenitis. These findings will assist in the development of improved protocols for the diagnosis of infectious diseases of dairy cattle"

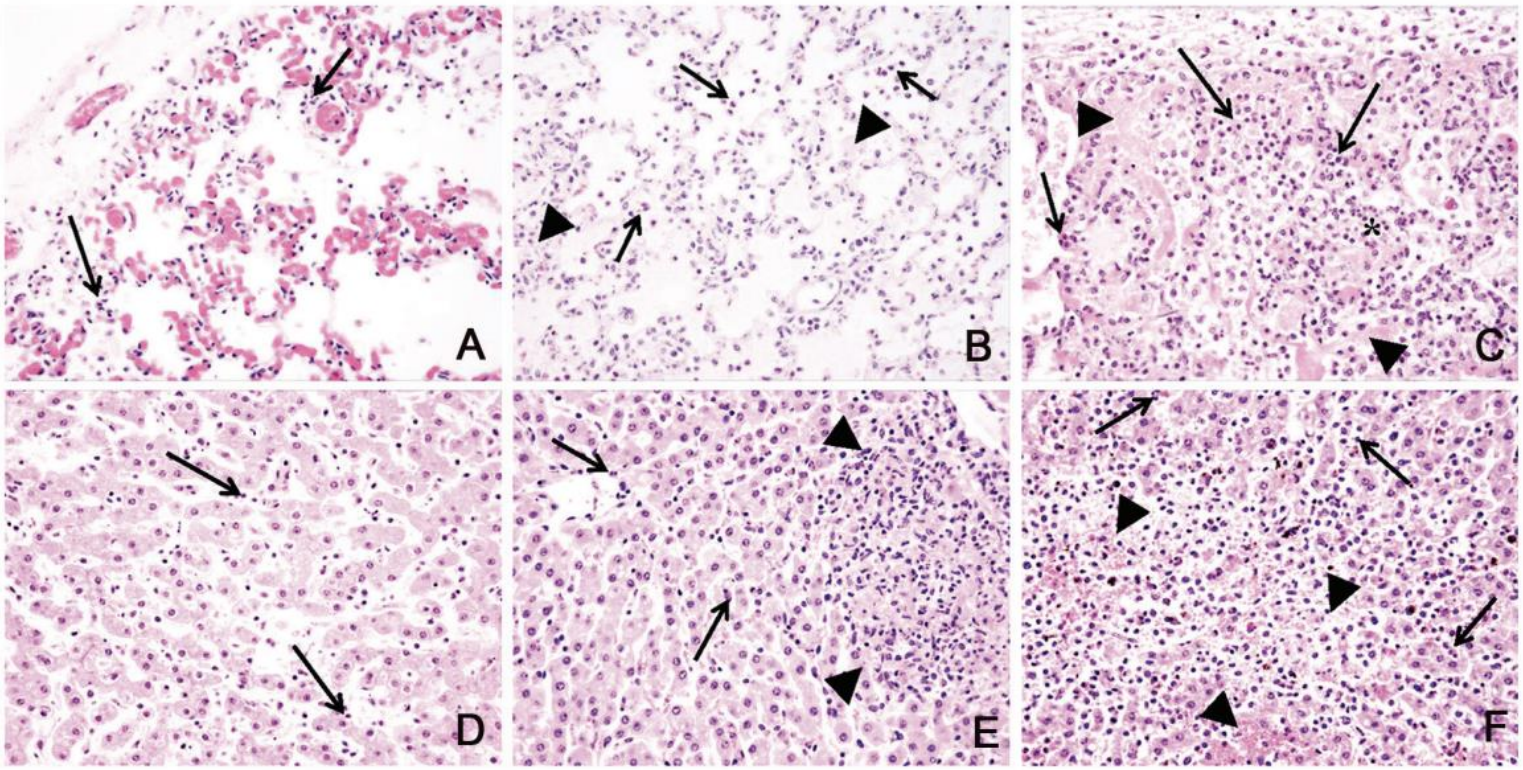


Figure 2. Representative photomicrographs of lung (upper panel) and liver (lower panel) from Holstein cattle <6 mo of age, infected with *Salmonella* Dublin. Alveolar capillaries with mild (**A**), moderate (**B**), and severe (**C**). circulating neutrophilia (arrows) together with occasional aggregates of fibrin within alveolar septa (B, arrowheads) or within alveolar spaces (C, arrowheads), and areas of necrosis characterized by karyorrhectic debris (C, asterisk). Sinusoids with mild (**D**), moderate (**E**), and severe (**F**) circulating neutrophilia (arrows) together with fibrin aggregates (E and F, arrowheads), and neutrophils and histiocytes surrounding necrotic hepatocytes forming paratyphoid granulomas (E and F). H&E.

Pecoraro et al., 2017

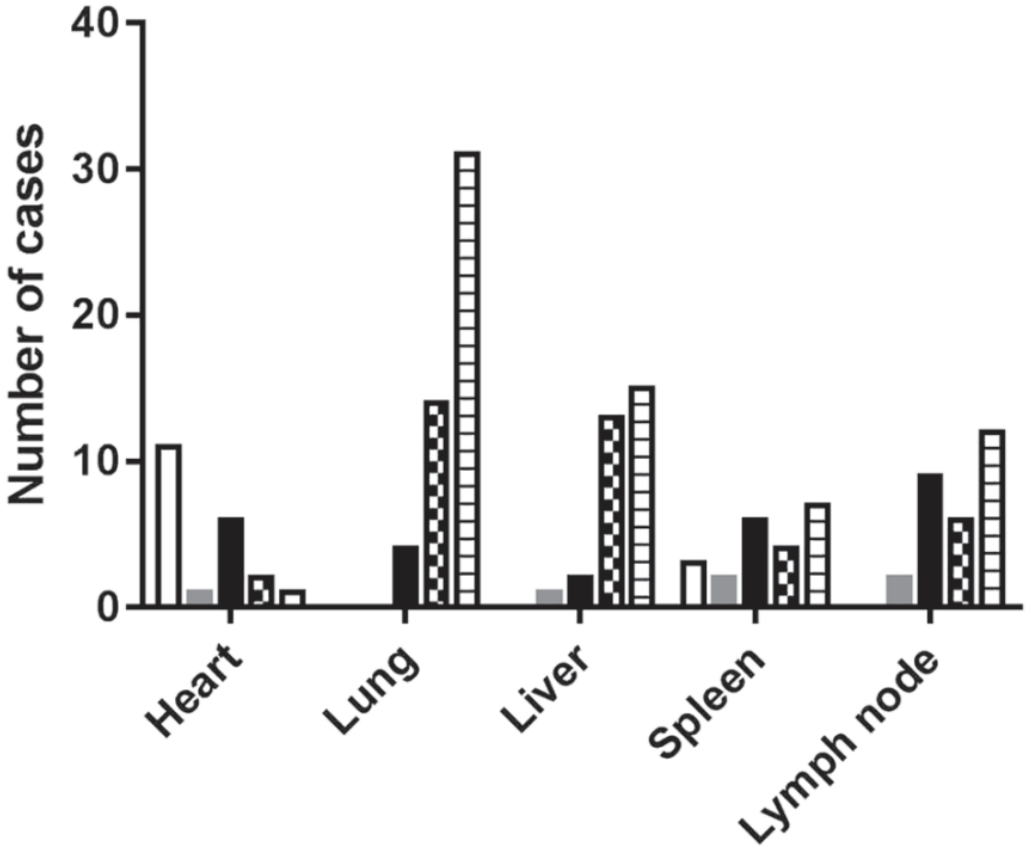


Figure 1. Severity of lesions based on the scoring scheme proposed in Table 1 for *Salmonella* Dublin culture-positive Holstein cattle <6 mo of age. Included is a total of 49 calves with sections of heart ($n = 21$), lung ($n = 49$), liver ($n = 31$), spleen ($n = 22$), and lymph node ($n = 29$) tissues submitted for histologic assessment. Key: no lesion (white), minimal change (gray), mild suppurative inflammation (black), moderate suppurative inflammation (checkers), severe suppurative inflammation (stripes).

Pecoraro et al., 2017

Diagnosis

- **Detection of the organism by culture or PCR**

  -- Feces (relatively low sensitivity in culture)

  -- Specimens from necropsy (recommended for culture)

  -- Blood in cases of suspected bacteremia

- **Necropsy** – No pathognomonic lesions

  -- Common: pulmonary congestion, suppurative pneumonia, catarrhal hemorrhagic enteritis, mesenteric lymphadenitis; and hepatosplenomegaly… → septicemia

- **ELISA** – monitoring

  -- Serum and milk

Prevention & Control

Goal : Minimize pathogen exposure & maximize pathogen resistance

Strategy

- Farm management
- Sanitation
  --Disinfectants can be used: halogens (eg, dilute chlorine bleach), phenols, quaternary ammonium compounds, and oxidizing agents (eg, potassium peroxymonosulfate).
- Biosecurity
  -- Rodent control

Both humeral & cell-mediated immunity are important!

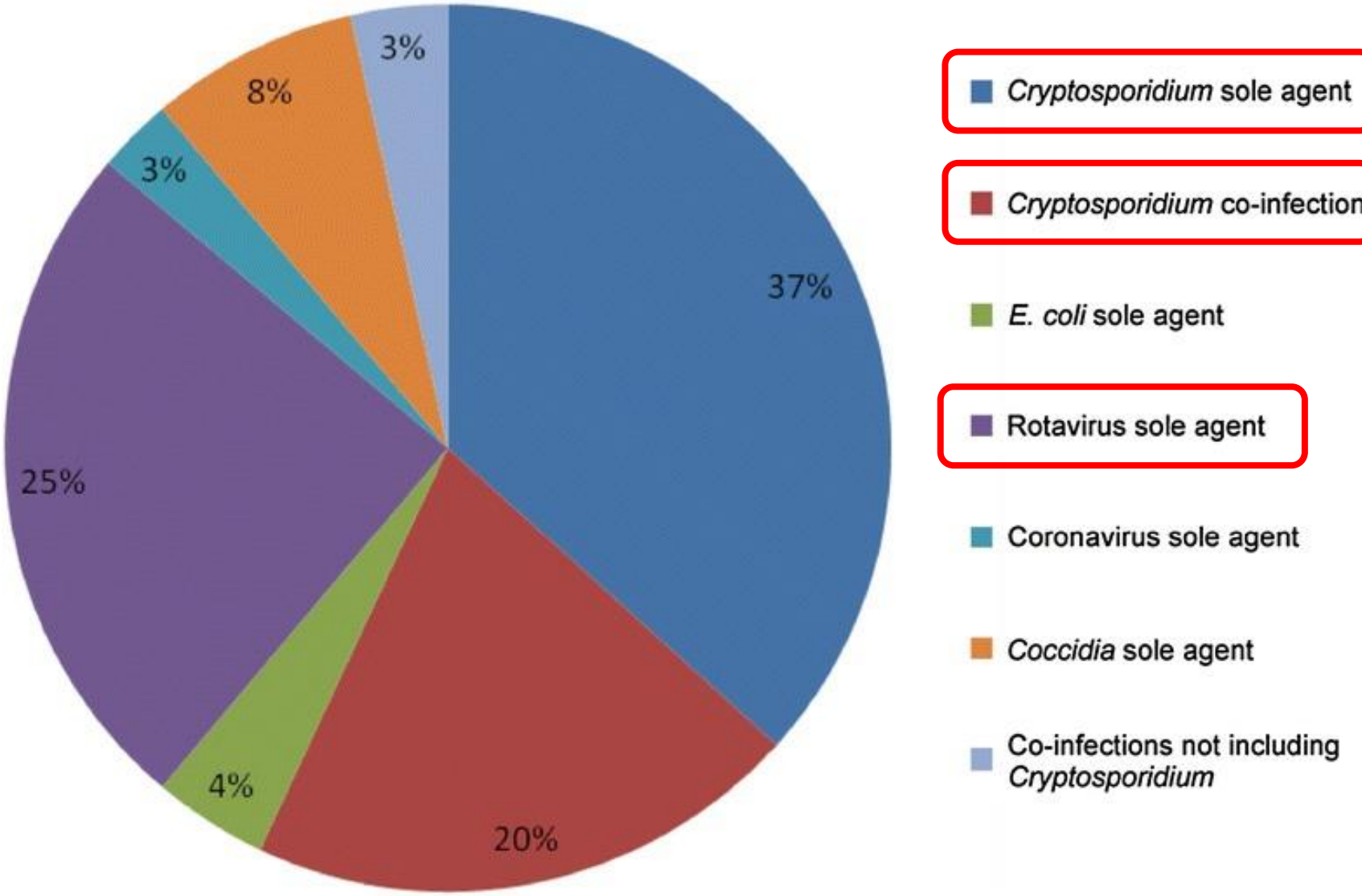
- **Vaccination**: **live, attenuated** whole-bacterium vaccines

Bovine Cryptosporidiosis

Bovine Cryptosporidiosis

- Worldwide distributed
- Many species and genotypes of Cryptosporidium. ***C. parvum*** (formerly C parvum type II) is **zoonotic** and infective to many animals, including people and calves. C. hominis (formerly C parvum type I) has a narrow host range and is transmitted primarily from person-to-person.
- Other cryptosporidial species have been isolated from cattle: *C andersoni*, *C bovis*, & *C ryanae*
- ***C parvum*** is a common cause of **calf diarrhea**, and its oocysts have been detected in the feces of 70% of 1- to 3-week-old dairy calves

小牛 C. parvum盛行率26.5 % (60/226)，成牛盛行率19.7 % (39/198)，場陽性率90 % (18/20)。——李睿彧，2018。



Pathogens causing diarrhoea in young calves. *Cryptosporidium* is the most commonly detected pathogen causing diarrhoea in calves less than 1 month of age as a proportion of diagnosable submissions 2007–2011 (veterinary investigation diagnosis analysis [VIDA]).

Thonsom et al., 2017

Cryptosporidiosis-密爾沃基事件

- **美國最嚴重的一次爆發案例。**
- **1993年4月威斯康辛州的密爾沃基(Milwaukee)，供水受到*C. parvum*的污染。事件爆發推測可能與上游農莊排水污染公共用水有關。大多數受感染的病人一到兩周內康復，但兒童、老人和免疫低下的人群可能出現嚴重的發病甚至死亡。密爾沃基事件之估計發病人數為403,000人，其中4,400人住院、幾個愛滋病人死亡，震驚全美。**

Slide provided by 鄭益謙老師

【隱孢子蟲】新州昆州逾千人感染腹瀉 當局發游泳警告

新州和昆州共錄得超過1,300人感染腹瀉疾病，衛生部門警告腹瀉患者避免到泳池游水至少兩星期，並敦促民眾在出現症狀時立即看醫生。



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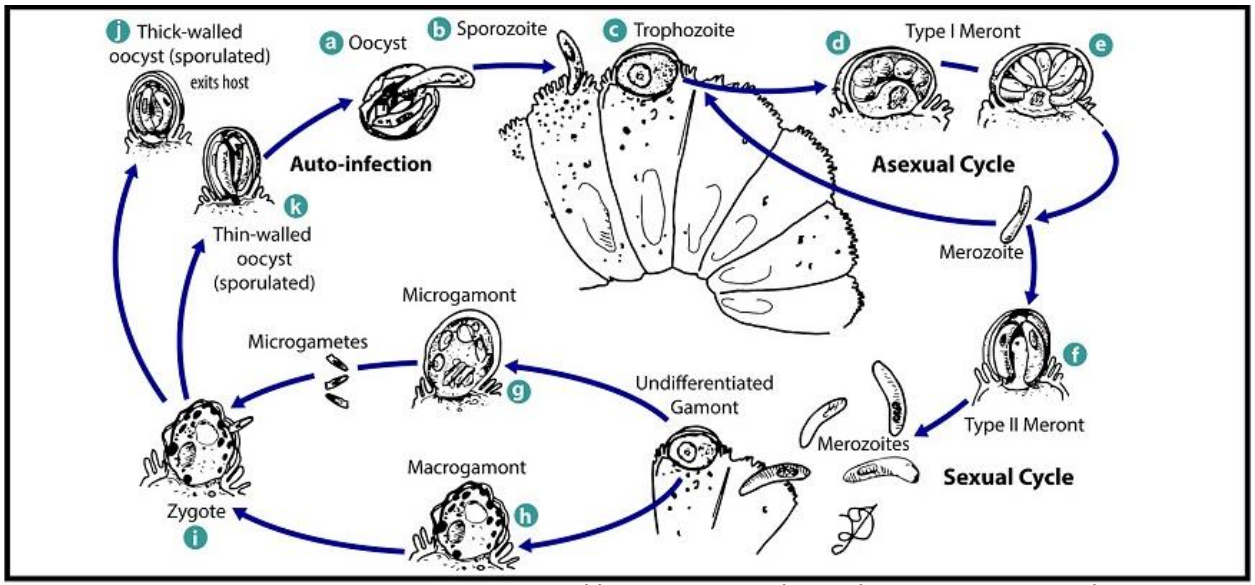
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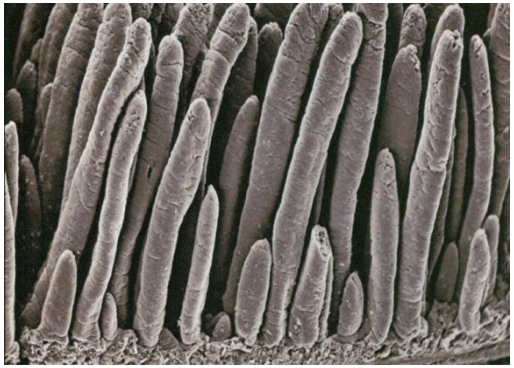
Etiology–*C parvum*

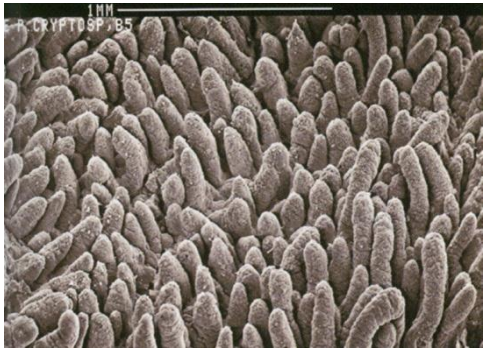
- No host specificity
- Completes its entire life cycle in a single host within 12 days.
- Oocysts are small (4-5 µm in diameter).
- Capable of survive in the **environment** for several months.
- **Resistant** to most disinfectants, including chlorine.
- Colonize at:
  --**GI tract**
  --bile duct
  --pancreas
  --respiratory tract
  --tonsils.



https://www.cdc.gov/dpdx/cryptosporidiosis/index.html

Cryptosporidium parvum



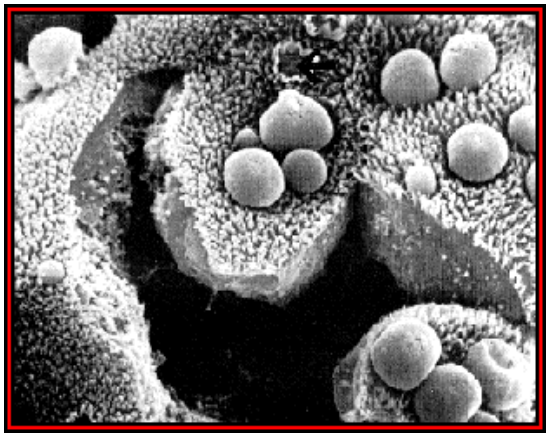


(Normal villi)

(From: **劉振軒等**, 2002)



Oocyst



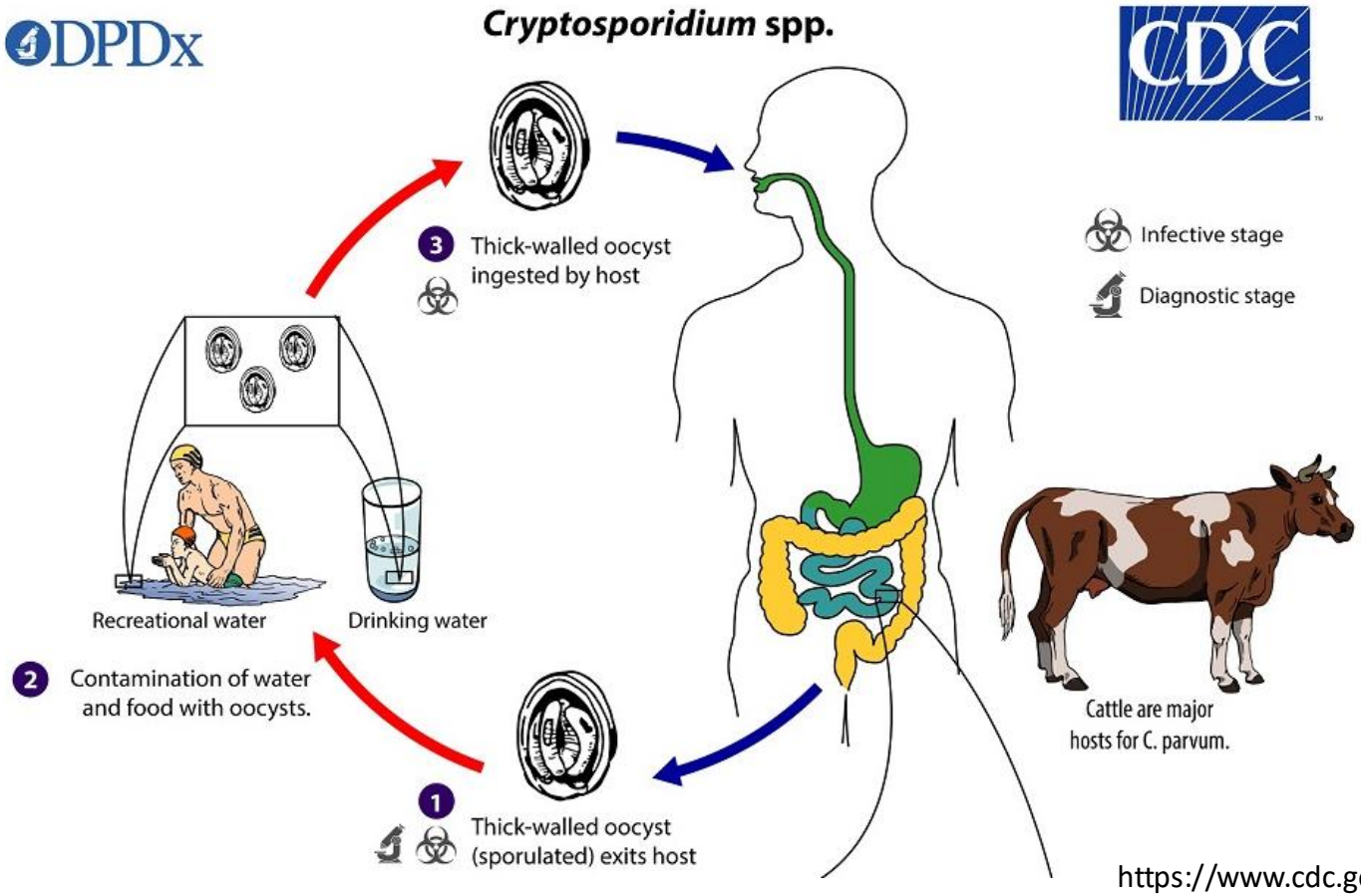
Oocysts on the intestinal lining

(From: Ifkovic, 2004)

Slide provided by 鄭益謙老師

Transmission

- Fecal-oral route (thick-walled oocysts)
  -- Ingestion of contaminated food or water



- Very few oocysts are required to cause infection in susceptible hosts
- As few as 5-17 oocysts were sufficient to cause diarroea and oocyst shedding

https://www.cdc.gov/dpdx/cryptosporidiosis/index.html

Clinical Presentation

- Vary, depending on immunity and ranging from asymptomatic (>6 weeks), **self-limiting** to sometimes life-threatening diarrhea.
- Calves aged 1-3 weeks are particularly susceptible, causing persistent **diarrhea**, **dehydration**, weight loss, and poor development.
- Infection can be complicated by secondary infections with other pathogens, such as rotavirus or coronavirus



Diarrheal fecal matting, calf
Diarrheal fecal matting on a neonatal Holstein calf with cryptosporidiosis.
Courtesy of Dr. William Harold Witola.

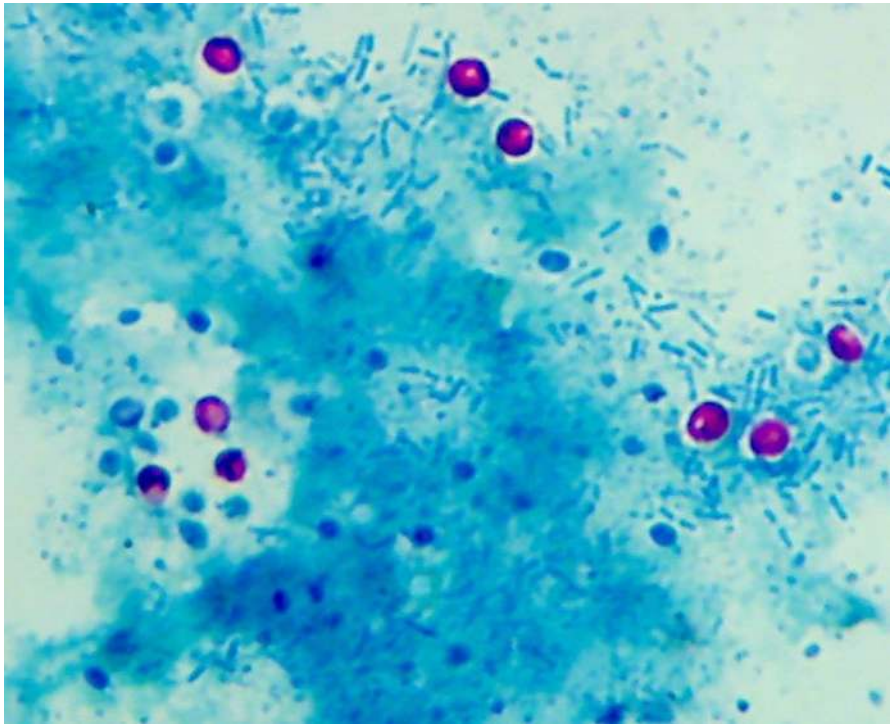


Cryptosporidial diarrhea, calf
Yellowish, watery, and mustard feces from a calf with cryptosporidiosis.
Courtesy of Dr. William Harold Witola.

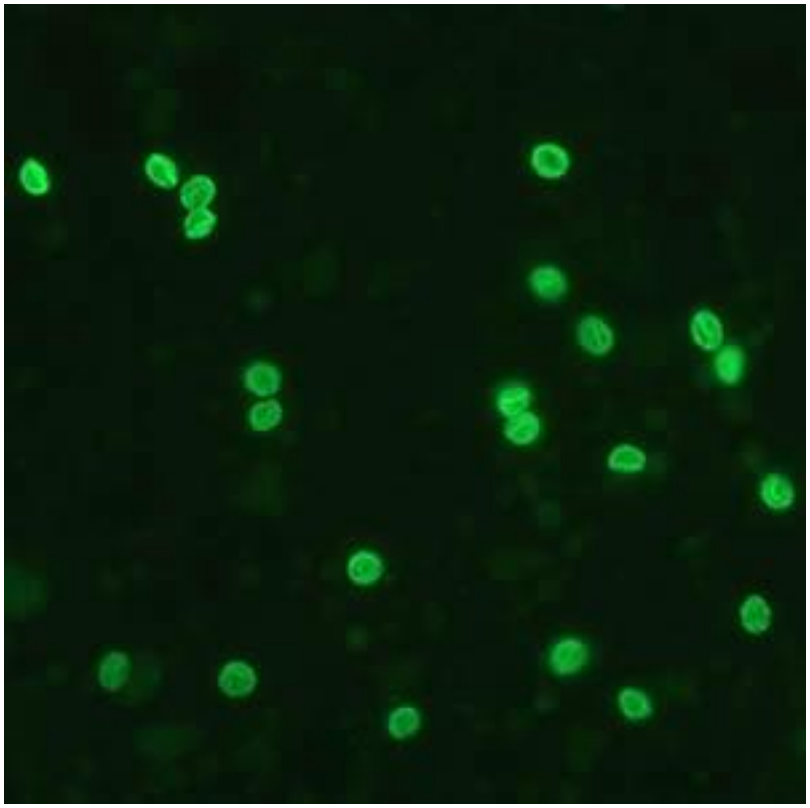
芥末醬...

Diagnosis

- Examination of **oocysts** in feces or drinking water.
  -- **Acid-fast** stain
  -- MAb immunofluorescence staining
  -- EIA
  -- PCR
- ELISA antibody detection
- Histopathological examination



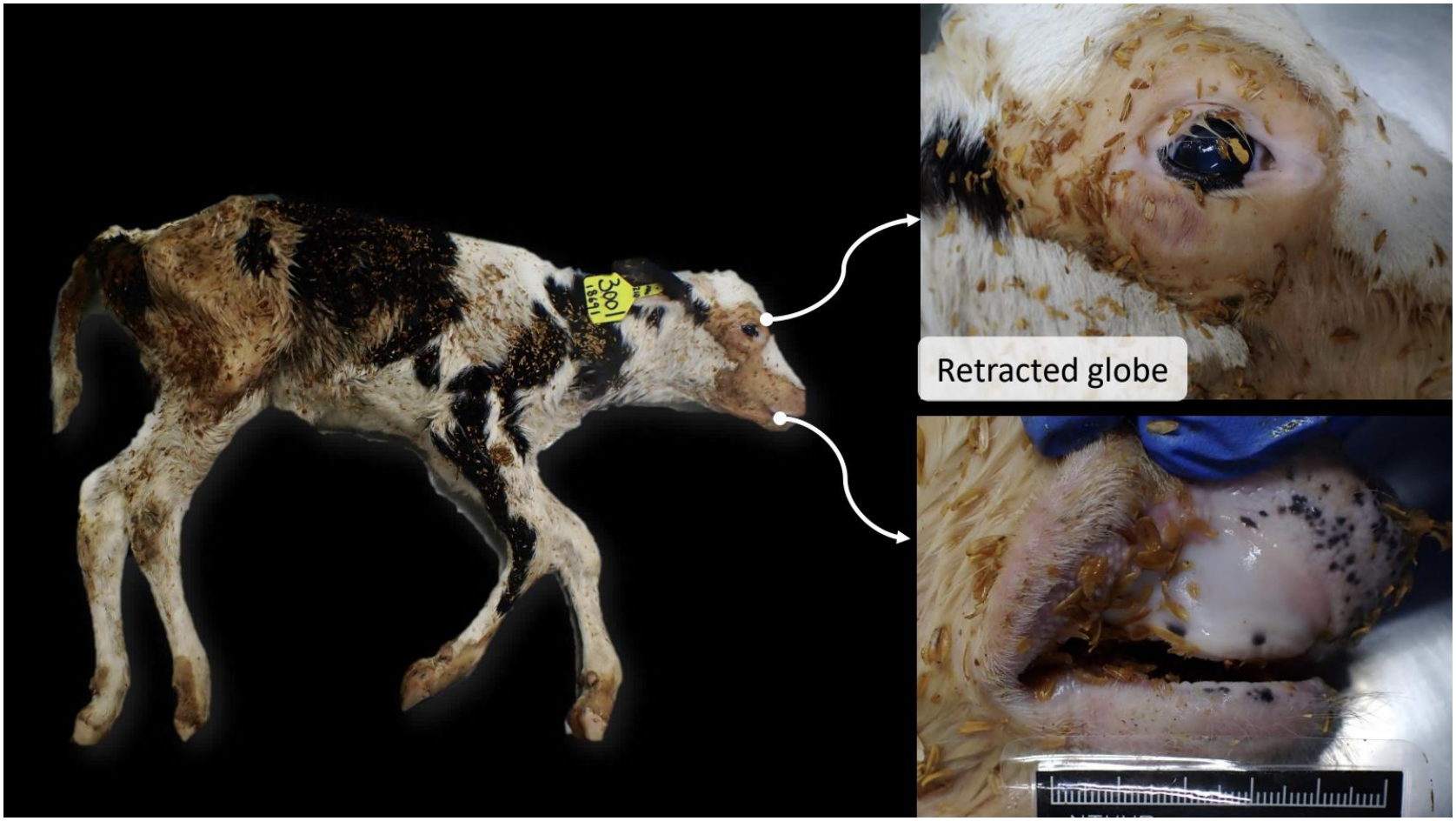
***Cryptosporidium parvum* oocysts, fecal smear**
Cryptosporidium parvum oocysts (red spherules) in a Ziehl-Neelsen–stained fecal smear.
Courtesy of Dr. William Harold Witola.

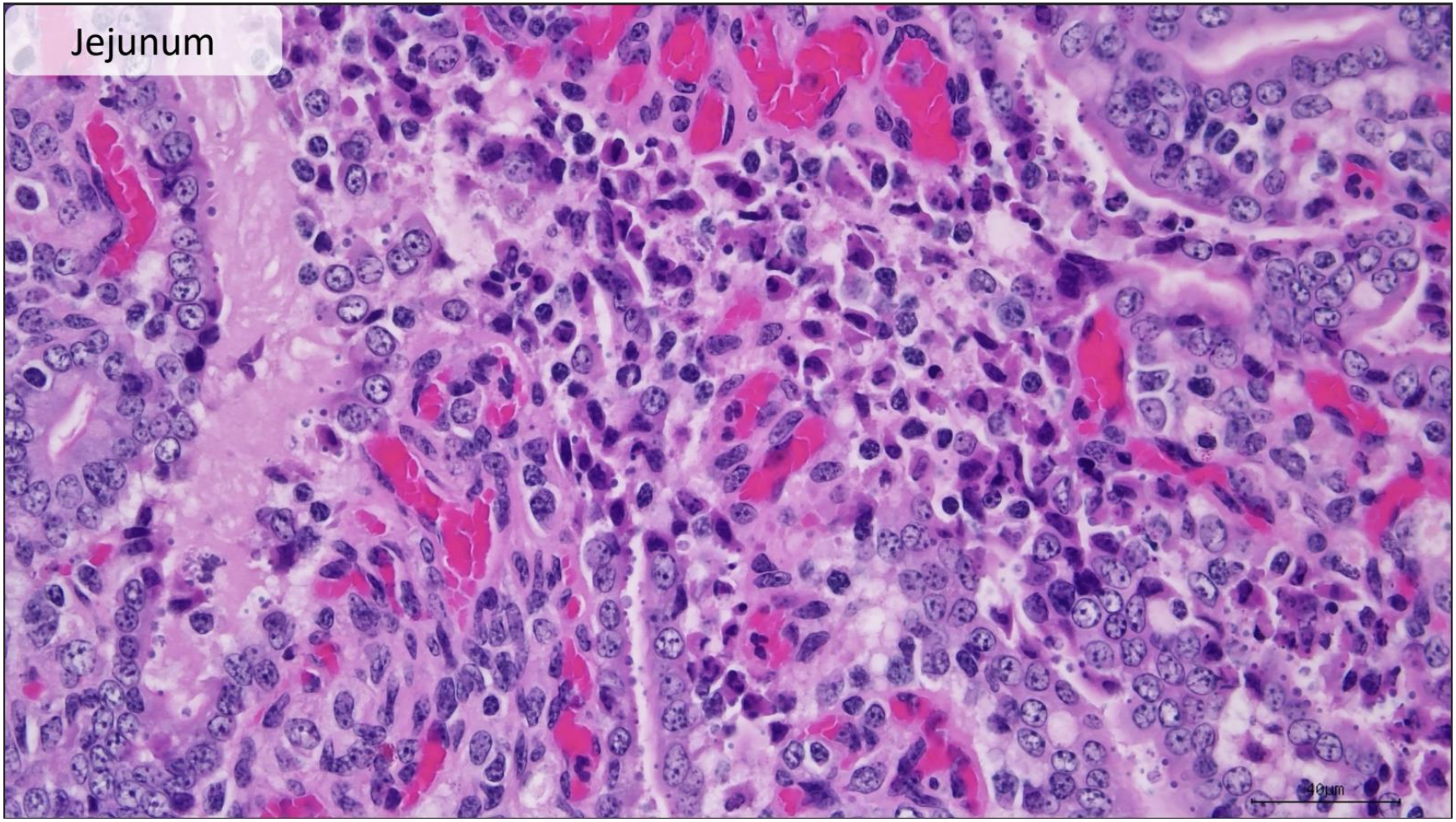


Cryptosporidium sp. oocysts labeled with immunofluorescent antibodies. Images contributed by the Kansas Department of Health and Environment.

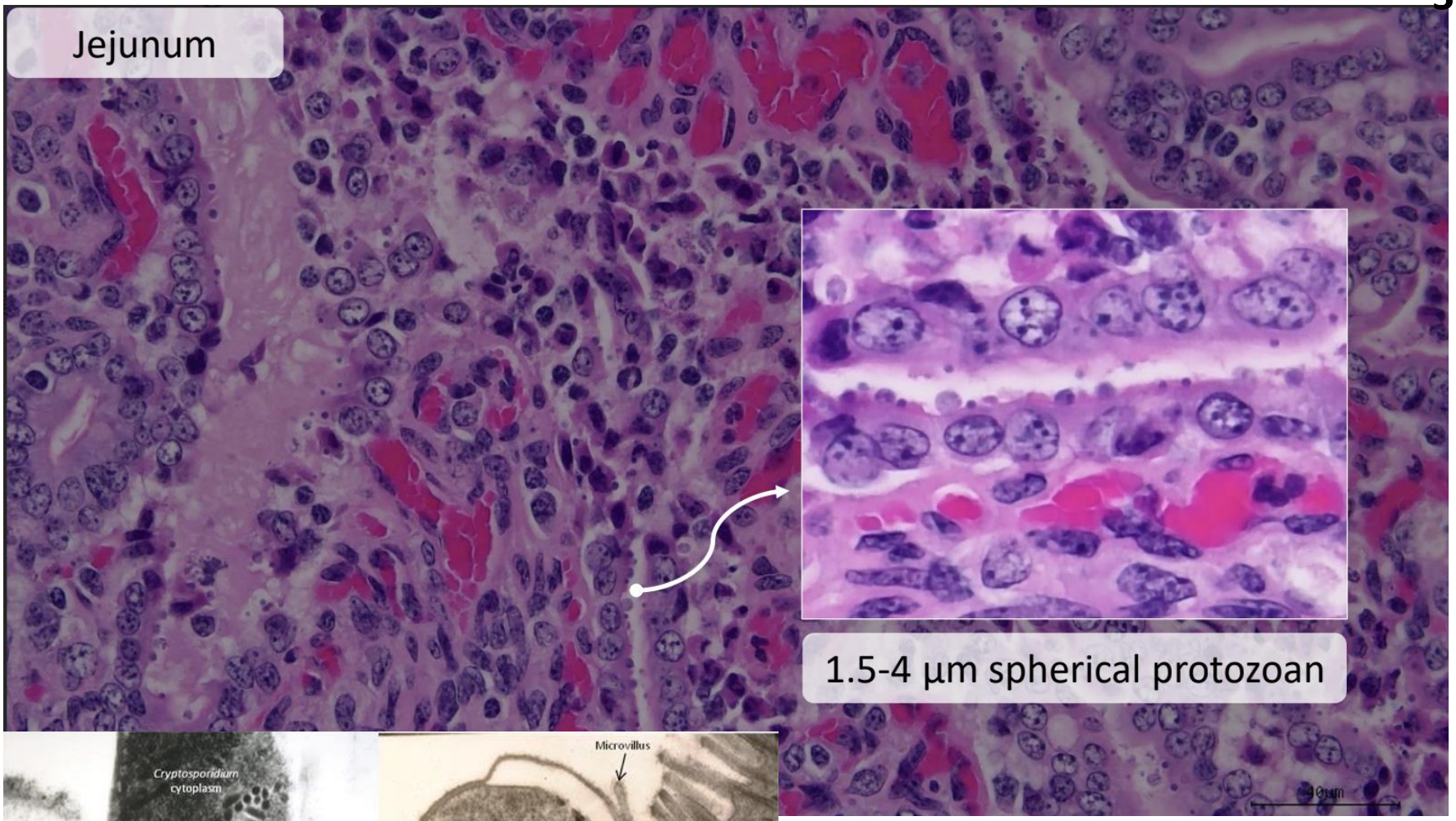
- **8-week-old calf**
- Farm condition:
  --Other calves (most): diarrhea
  --Mother cow: no specific symptoms

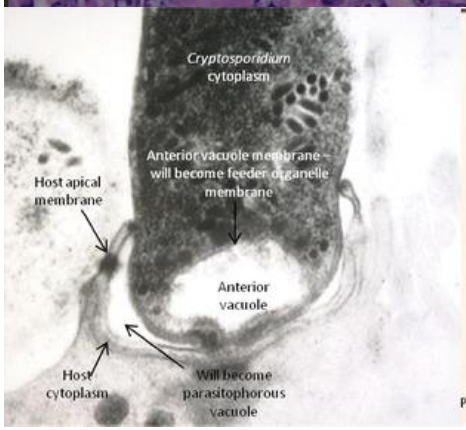


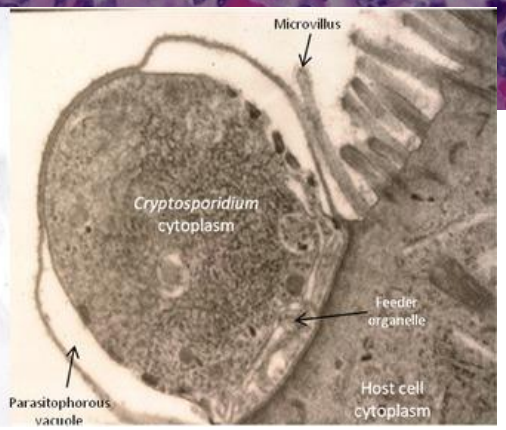

Retracted globe



Numerous small round blue **organisms** colonizes the apical borders of enterocytes.



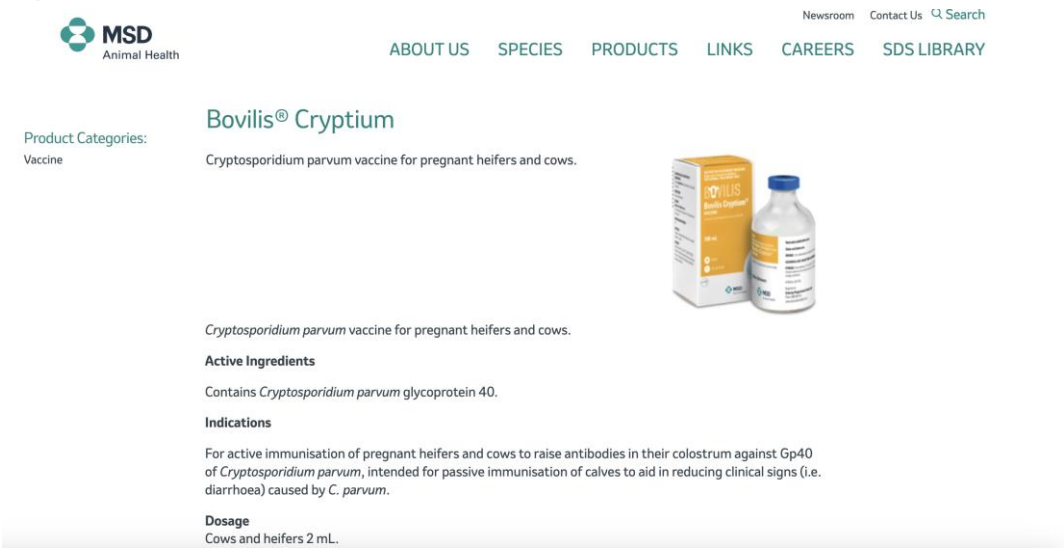




Thonsom et al., 2017

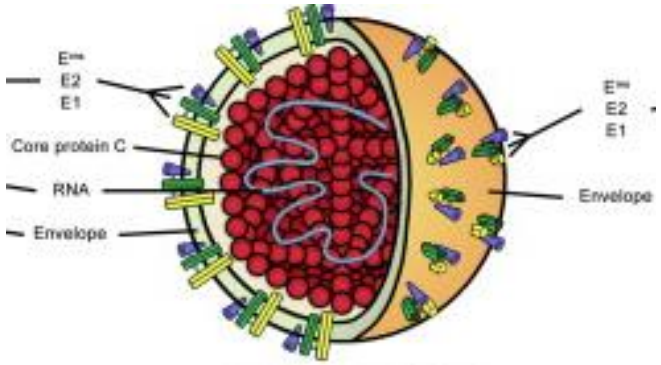
Prevention & Control

- Hygiene and reduction of environmental oocyst burden
- Adequate colostrum intake
- Fluid/electrolyte supportive therapy
- Anticryptosporidial drugs have limited efficacy
- **Maternal vaccination against C. parvum is now available in some regions**



Bovine Viral Diarrhea

Etiology



- Bovine viral diarrhea virus (BVDV)
- Three distinct species:
  --BVDV-1 (*Pestivirus bovis*),
  --BVDV-2 (*Pestivirus tauri*),
  --HoBi-like virus or BVDV-3 (*Pestivirus brazilense*)
- Genus *Pestivirus,* family *Flaviviridae*
- Two biotypes: cytopathic (CP) & non-cytopathic (NCP)

  **** The biotype does not reflect the ability to cause disease**
- Wide host range: cattle, sheep, swine, goats, and other wild animals
- **Immunosuppression**

- Widespread in cattle population due **to persistently infected (PI) animals** associated with prenatal infection (during the **first 4 months** of fetal development) with **NCP** virus strains.
- The prevalence of infection vary depending on the vaccination practices, population densities and housing as well as management.
- In the field, the prevalence of BVDV-1b increases (43% → 75–100%) after 2001; the usage with 1a & 2a strain-based vaccine likely contributed to this shift.
- Economic significance of BVDV is related to **reproductive failure**, impaired calf health, increased risk of infectious disease, reduced milk production and mortality losses...

Transmission

- **The major source of infection is the PI viremic animals.**
- The estimated mean animal prevalence PI in a herd is about 1%–2%.
- BVDV is present in all secretion and excretion.
- Route of transmission:
  - ✓ Direct contact: nose-to-nose
  - ✓ Transplacental
  - ✓ Indirect contact: airborne, flies, fomites…
  - ✓ Artificial insemination/embryo transfer
  - ✓ Contaminated vaccination (from FCS used in TC)

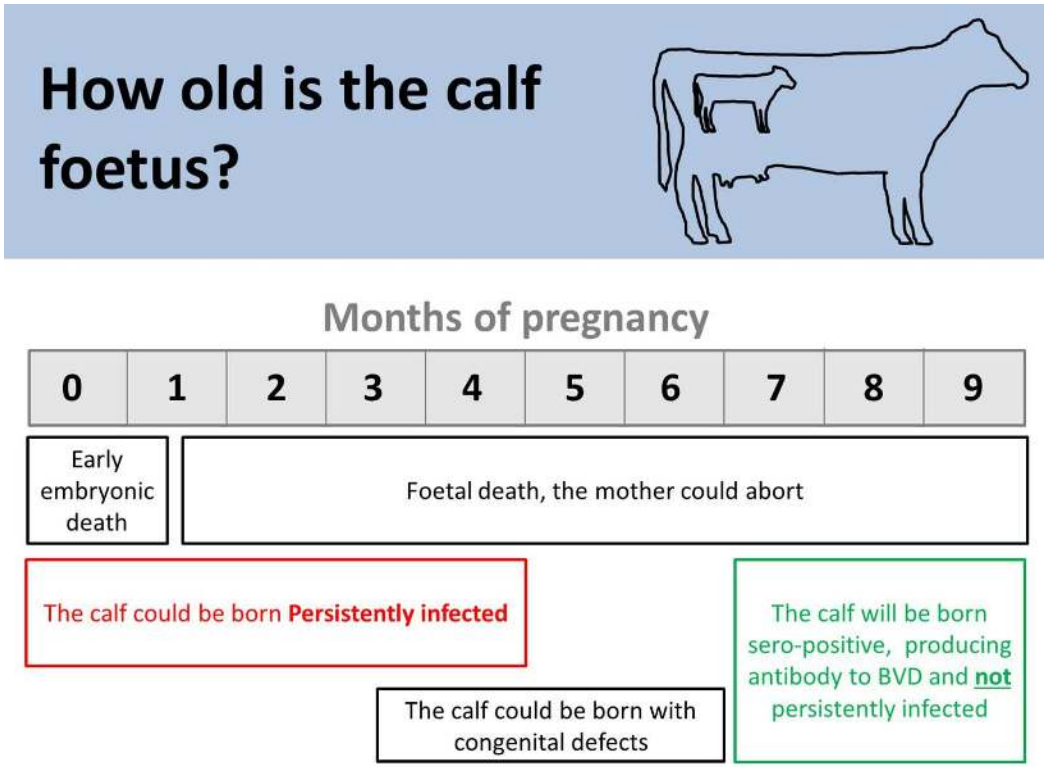
Pathogenesis & Immunity

- The virus infects **cells of the innate immune system** affecting the function of neutrophils, monocytes, macrophages, and dendritic cells.
  - → **Immunosuppression**
  - → BVDV may also participate in the development of BRD
- Glycoprotein **E2** is the major antigenic target for antibodies
- Cattle respond to **natural infections** or vaccination with **MLV** vaccines exhibit **a long-lasting immunity**
- High levels of VN Ab prevent disease following **homologous** challenge but animals may still develop viremia and shed virus in nasal secretion.

Clinical Presentation

- **Vary,** depending on age, immune status, viral virulence, stage of gestation…
- Fetal infection
  - Reproductive failure (abortion, mummification, stillbirth…)
  - Congenital abnormalities (microcephaly, hydrancephaly, cataract, osteopetrosis, pulmonary hypoplasia…)
  - **Persistent infection (PI) – major source of herd infection!**
- Classical/acute BVD
- **Mucosal disease**
- Severe acute BVD

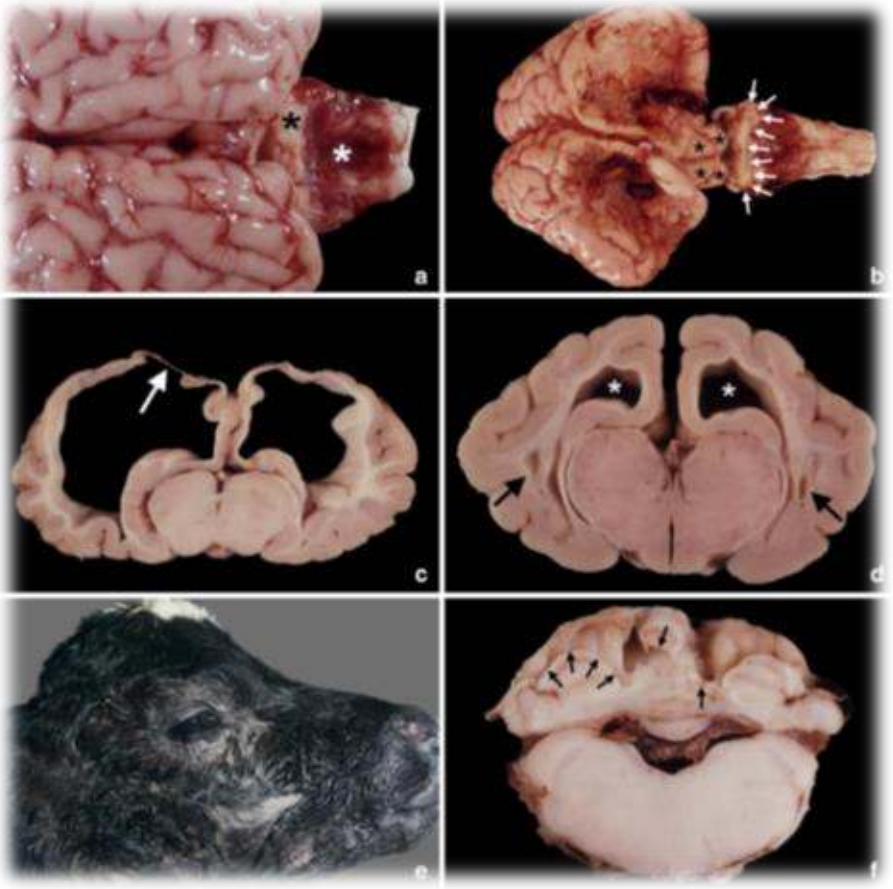
Outcomes of Fetal Infection



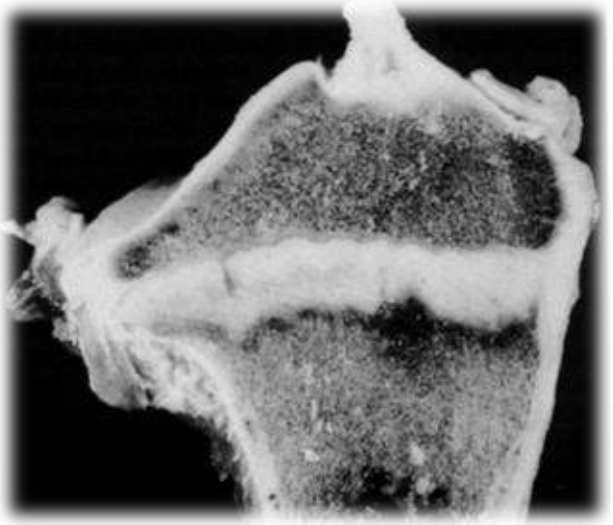
- **NCP** infection **prior** **to 120** days gestation:
  - Fetal death, abortion, mummification
  - **Persistent infection (PI)**
- NCP infection **during 90-150** days of gestation:
  - Congenital defects

Although both biotypes can cross the placenta, to invade the fetus, only NCP virus strains can establish persistent infection in the fetus.

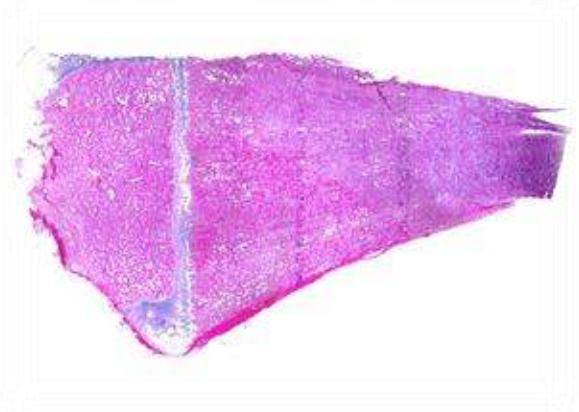
Congenital anomaly of CNS

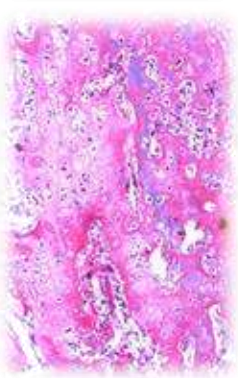


Deranged osteogenesis









Agerholm et al., 2015

Classical/Acute BVD

- Usually associated with NCP strains
- Transient infection in immunocompetent cattle over 6m/o
- Clinical signs are **mostly inapparent to subclinical**; if present, include fever, mild oculonasal discharge, tachypnea, diarrhea…
- Transient **leukopenia** → secondary infection
- **High morbidity, low mortality**
- Lifelong immunity

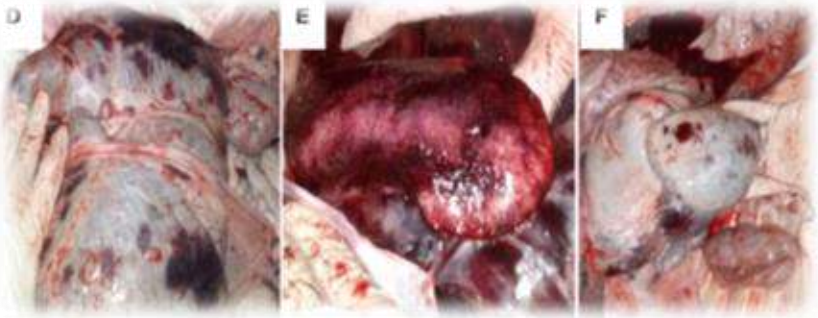
Severe Acute BVD

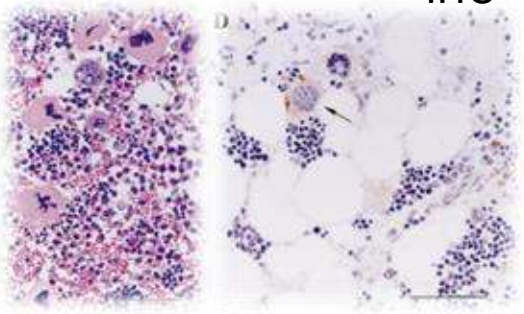
- **High morbidity, high (25%) mortality**
- All animals at all ages are susceptible
- Usually due to **NCP BVDV-2**
- Clinically indistinguishable from mucosal disease (MD) and signs may include: high fever (up to 42 °C), respiratory distress, anorexia, profound watery diarrhea & agalactia
- **Thrombocytopenic/hemorrhagic syndrome**
- PI calves are commonly born several months following the outbreaks.

Apoptotic megakaryocytes

IHC





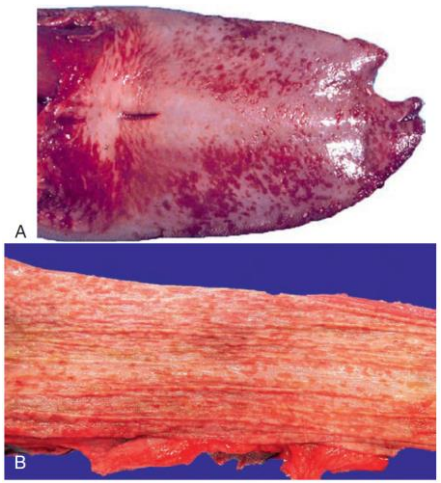


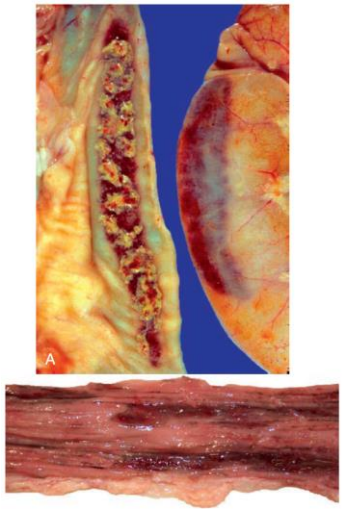
Mucosal Disease (MD)

Internal or external

- PI animals infected postnatally with CP strain
- **Moderate morbidity, high (100%) mortality**
- Usually during 6 m/o to 2 y/o
- Clinical signs include high fever, leukopenia, watery diarrhea, inappetence, dehydration, erosive lesions of the nares and mouth, and death within a few days of onset.
- Acute MD vs chronic MD (eventually will die)

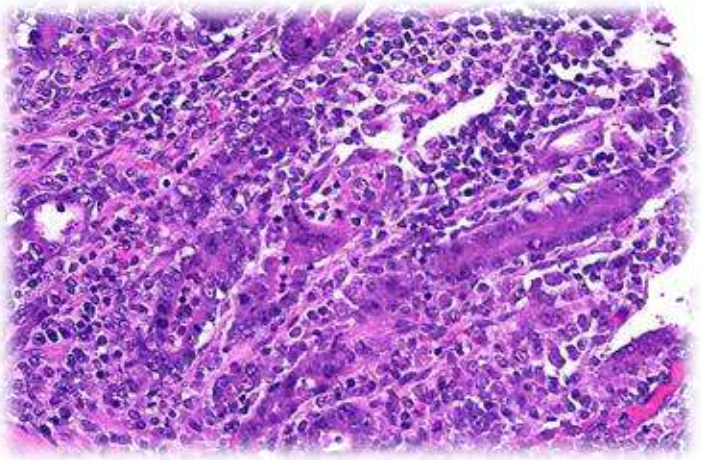


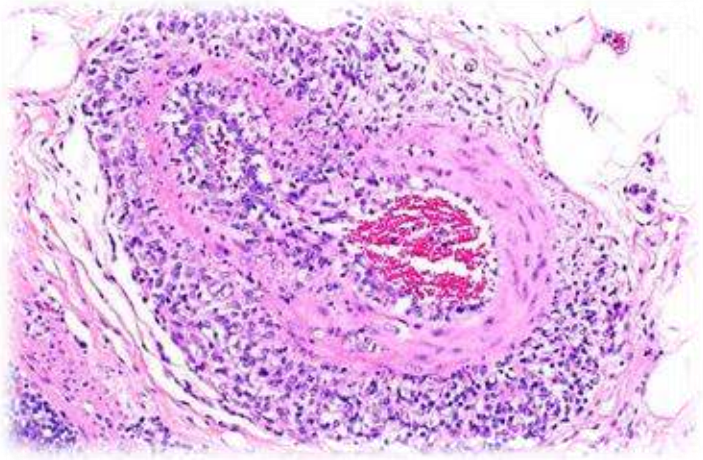






Rinderpest	Malignant Catarhal Fever
***Paramyxoviridae*, morbillivirus**	***Herpesviridae*, gammaherpesvirus**
Officially declared **eradicated** !!	
Fever, oral erosions, diarrhea, **lymphoid necrosis**, and high morbidity and mortality	**Lymphoproliferation**, vasculitis, and erosive to ulcerative mucosal lesions
• **Erosions, ulcerations**, and hemorrhagic lesions affecting the oral cavity • Hemorrhagic, necrotic, edematous Peyer's patches • Hemorrhage and congestion of cecum, colon, rectum (zebra stripes)	• Lymph nodes: Enlarged, edematous (common in all species, except bison) • **Mucosal surfaces: Hyperemia and edema, erosions and ulcerations** • Organomegaly due to lymphoproliferative vasculitis
• Lymphoid necrosis with loss of mature lymphocytes and replacement by plasma cells and macrophages • Necrosis of intestinal crypts with resultant erosions and ulcers • **Syncytia** (oral mucosal epithelium); **Intracytoplasmic and intranuclear eosinophilic inclusion bodies** in infected epithelial cells and lymphoid cells • Intestinal lesions most severe over Peyer's patches	• Widespread arteritis-phlebitis of medium caliber blood vessels • **Lymphoid proliferation and production of large lymphocytes** • Mucosal ulceration in digestive, urinary, and respiratory tracts





Prevention & Control

- **Detection** and **elimination of PI animals** from the herd.
- Prevention of introduction of infection into herd.
- Vaccination of breeding females to **prevent fetal infection**.
- Both inactivated and MLV vaccines are available; however, antigenic diversity among BVDV may affect the efficacy of a given vaccine
- **MLV vaccine is not recommended** to be used in pregnant cattle.
- Eradication can be done by detection and elimination of PI animals, avoidance of vaccination, and strict biosecurity measures to prevent introduction of PI animals into the herd.

Winter Dysentery of Cattle

Etiology

Both GI and respiratory tropism
Small & large intestine

- Bovine **coronavirus** (BCoV)
- BCoV not only causes diarrhea in adult cattle (winter dysentery, WD) but also in calve (calf diarrhea, CD).
- BCoV can also induce respiratory disease in calve and adult cattle.
- Ubiquitous in the cattle population, persisting in subclinically infected animals.
- High morbidity (30-50%) with outbreaks but low mortality.
- Transmission by fecal–oral route; feces and nasal secretion are the primary source of infection
- One serotype

Clinical Presentation

- Incubation period: 3–7 days
- Explosive outbreak of diarrhea preceded by transient fever
- The feces are liquid, often **hemorrhagic** and homogeneous without much odor
- **Markedly reduced milk production** that lasts for up to one week
- Respiratory signs may occur: cough, nasolacrimal secretion
- Death is usually associated with severe blood loss.

Prevention & Control

- No vaccine is commercially available
- No specific control measures available.
- Hygiene.
- Minimize overcrowding in dairy housing.

Bovine Rotavirus

Etiology

- Genus *Rotavirus* in the family *Reoviridae*

Potential of reassortment

- No-enveloped virion possessing 11 dsRNA **segments**
- The most recognized pathogen causing **acute diarrhea** in calves **under one month of age**
- Seven serogroups
- Most BRVs (95%) belong to **group A**, although groups B and C rotaviruses have also been identified in field cases
- Also has been recognized as the major pathogen of acute diarrhea in both humans and animals.

→ Both **zoonotic** and economic importance

Pathogenesis

- Transmitted by **fecal-oral route** though potential airborne and waterborne transmissions are also reported
- Rotavirus infects the **mature** nondividing **enterocytes** in the middle and top parts of the villi in the **small intestine**
- Rupture and sloughing of the enterocytes with release of virus to infect adjacent cells

Clinical Presentation

- Disease severity depends on the intake of colostrum, age and health status, immune status of the dam, degree of exposure and virulence of virus, and the presence of secondary pathogens
- **Short incubation time** (12–36h)
- It may also occur in calves within 24 hours of birth.
- Calves between 1 day and 3 weeks old, especially those between **7 and 10 days old are most susceptible**.
- Clinical signs include acute onset of diarrhea, dehydration, weakness and sunken eyes but low mortality in uncomplicated cases.
- Co-infection with *E. coli* leads to much more severe diarrhea and dehydration and the affected calves may die within 24h.





(From:馮翰鵬, 2002)

Control & Prevention

- Rotaviruses are relatively **resistant** to inactivation by chemical disinfectants and antiseptics.
- Control and prevention measures against rotavirus infection are not easy for its mass distribution and tendency to stability in different climate situation and high concentrations shed in feces.
- The primary strategy to reduce the burden of rotavirus infections is **vaccination**.
- Both killed and MLV vaccines are available

Respiratory Diseases

Bovine Respiratory Disease (BRD)

Bovine Respiratory Disease (BRD)

- Also known as **shipping fever,** undifferentiated fever and enzootic calf pneumonia in young dairy calves
- A complex respiratory disease of a **multifactorial etiology** characterized clinically by dyspnea, coughing, nasal discharge, varying degrees of depression, anorexia, pyrexia ranging from 40 to 41° C (**high fever**).
- All ages are affected but **young**, rapidly growing cattle, especially recently weaned calves are most susceptible.

Bovine Respiratory Disease (BRD)

- BRD is most commonly associated with the **transport** and assembly of large groups **of recently weaned calves** into feedlots.
- Morbidity (up to 50 %) and case fatality (up to 10%) strongly depend on the array of risk factors present in the farm.

Etiology

Viruses

Bovine herpesvirus-1 (BHV-1) = infectious bovine rhinotracheitis (IBR)
Bovine respiratory syncytial virus (BRSV)
Parainfluenza-3 virus (PI-3V)
Bovine virus diarrhea virus (BVDV)
Bovine coronavirus (BoCV)
Bovine adenovirus (BAV)

Bacteria

Mannheimia haemolytica
Pasteurella multocida
Histophilus somni
Mycoplasma spp.
(Mycoplasma bovis)

Other

Environment (**weaning, transport**, commingling, crowding, temperature, dust, ventilation…)
Immunity
Vaccination program
Management
→ **Stress**

Bovine herpesvirus-1 (BHV-1)

- Also known as **infectious bovine rhinotracheitis** (IBR) virus
- Widespread in cattle population

  * In Taiwan (1997–1998): 71.6% (2003/2796) of the cattle tested were considered IBR antibody positive, and 11.6% (325/2796) were regarded as suspect

- One serotype, several subtypes:

  -- BHV-1.1 = respiratory subtype

  -- BHV-1.2 = genital subtype

  -- ~~BHV-1.3 = encephalitic subtype~~

  →Reclassified as a distinct genotype BHV-5

Various clinical diseases depend more on **the route of infection** than on the subtypes

- Transmission via contact with BHV-1-containing secretion/aerosol

Morbidity & Mortality

- Morbidity and mortality are **higher in feedlot cattle** than in dairy herds because of the frequent introduction of susceptible animals into an enzootic situation.
- Morbidity: 8% in diary cattle and 20–30% in feedlot cattle; may rarely reach 100% in unvaccinated cattle
- Mortality: 3% in diary cattle and variable in feedlot cattle depending on the severity of secondary bacterial infection.
- Newborn calves with systemic infection: 100%

Clinical Presentation

- Many:
  - ✓**Respiratory diseases** (IBR & predisposition to pneumonia)
  - ✓Conjunctivitis (may or may not concur with IBR)
  - ✓Genital diseases (Infectious pustular vulvovaginitis, balanoposthitis)
  - ✓**Reproductive failure & abortion**
  - ✓Encephalitis
  - ✓Mastitis
  - ✓Systemic disease in newborn calves
  - ✓Latency



5.3. IBR with purulent oculonasal discharge

Clinical Presentation

- The severity depend on the strain of the virus, the age susceptibility, and environmental factors.
- Clinical disease is **most common after 6 months** of age in colostrum-fed calves as a result of waning colostral IgG (passive immunity).
- Clinical signs: high fever (up to 42 ˚C), anorexia, coughing, excessive salivation, **nasal discharge** (sometimes accompanied by ulcers in mouth and nose), conjunctivitis with lacrimal discharge, inflamed nares (**red nose**), and dyspnea if the larynx becomes occluded with purulent material.

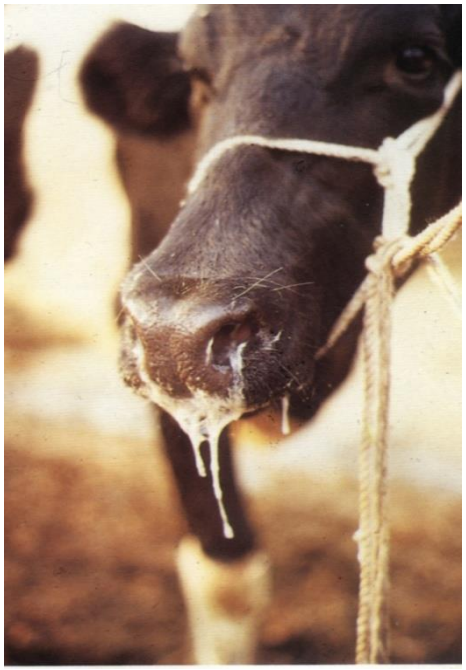
Predisposing to Pneumonia

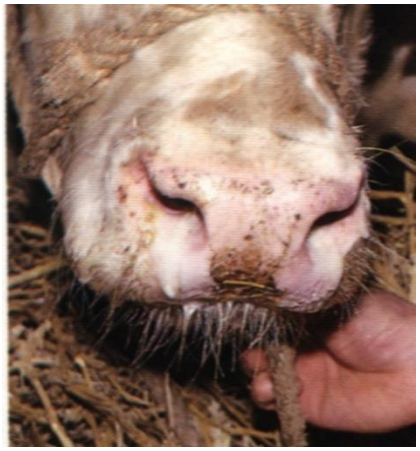
- BHV-1 virus **impairs the function of alveolar macrophages**, allowing *Mannheimia haemolytica* to persist and proliferate in the lung.
- BHV-1 affect the composition of alveolar phospholipids, which can alter the function of lung surfactant and compromise pulmonary defense mechanisms.
- BHV-1 infection affects bovine peripheral blood neutrophils, **enhances the binding of *M. haemolytica* leukotoxin** (the principle virulence factor) to bronchoalveolar leukocytes, and increases their killing.

The net effect is that BHV-1 infection amplifies the detrimental effect of *M. haemolytica* in the lung

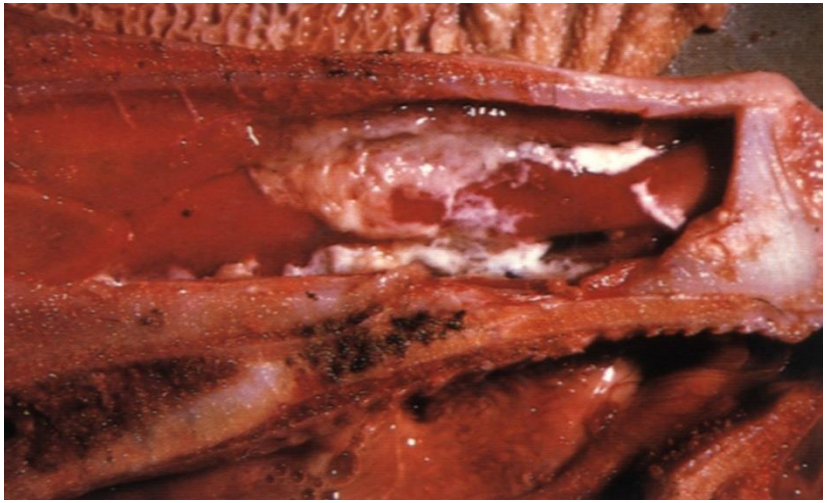
Reproductive Failure & Abortion

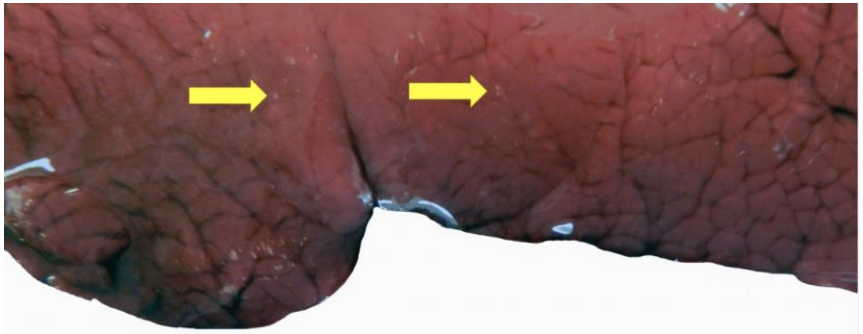
- A common sequela and occurs some **weeks after the clinical illness or parenteral MLV vaccination** of nonimmune pregnant cows.
- Abortion is most common in cows that are **6 to 8 months** pregnant.
- Abortion rate is usually low but reach 60% in nonimmune herd.
- Mechanism: peripheral blood WBC carry BHV-1→ placenta → fetus → abortion (within 48h)
- Endometritis, poor conception, and short estrus can occur after insemination with infected semen.











Aborted fetuses show moderately severe autolysis and focal necrotizing hepatitis.

https://www.vdl.ndsu.edu/abortion-workup-case-study/

(Blowey and Weaver, 1991)

Prevention & Control

- The vaccine will **prevent abortions** caused by the virus and provide **protection against respiratory disease** if given at least 10 days before natural exposure.
- Several live-attenuated and inactivated BHV-1 vaccines are currently available.
- The **parenteral MLV vaccine is potentially abortigenic** and cannot be used on nonimmune pregnant cattle.
- DIVA strategy has been applied in field to help eradicate IBR in some European countries.
- Antimicrobial therapy to prevent 2nd bacterial infection.

Table 1. Infectious bovine rhinotracheitis (IBR) *marker* vaccines available in European countries.

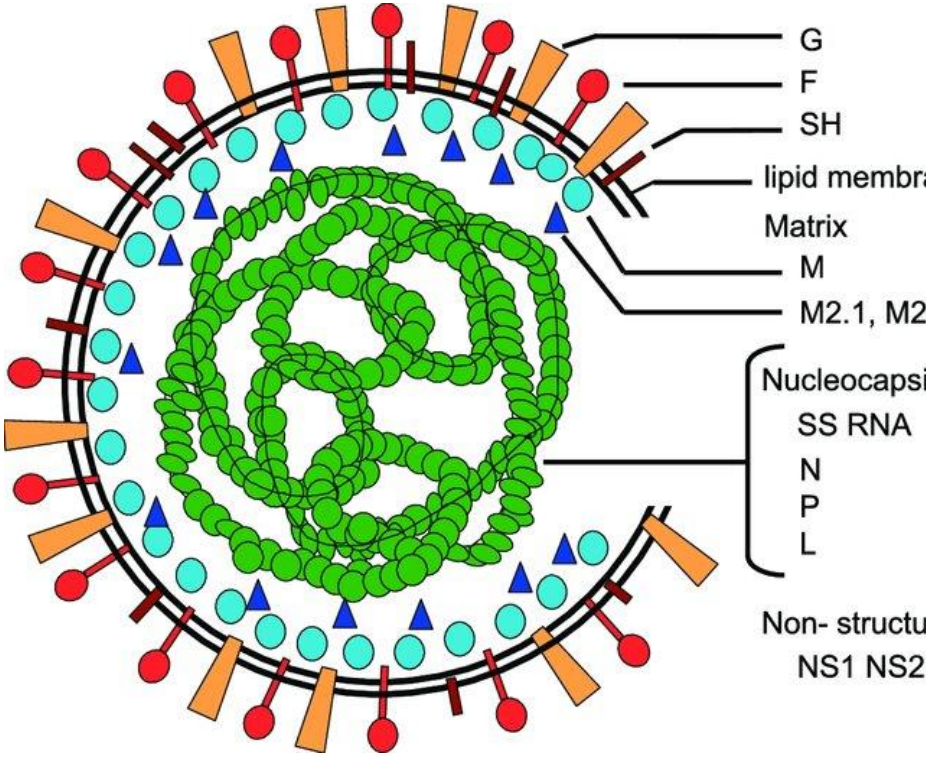
Name of Vaccine (Company)	Active Substance	Vaccine Strain	Dose	Route	Marketing Authorization Numbers
Hiprabovis IBR Marker Live (Hipra)	Live gE-, tk-, double-gene deleted BoHV-1 virus	Ceddel $10^{6.3}$–$10^{7.30}$ $CCID_{50}$ [a]	Single 2 mL	i.m.	EMEA/V/C/000158
Cattlemarker IBR Inactivated (Zoetis)	gE-, inactivated virus	Difivac gE-, ≥ 5.5 $\log_2$ [b]	Single 2 mL	s.c.	EMEA/V/A/115
Bayovac IBR Marker Vivum (Bayer)	gE -, modified live (attenuated) virus	Divifac 10^5 $TCID_{50}$ (min)–10^7 $TCID_{50}$ (max) [c]	Single 2 mL	i.n., i.m.	EMEA/V/A/023b/001
Bovalto Ibraxion Inactivated IBR virus (Merial)	gE-, inactivated IBR virus	0.75 VN.U [d]	Single 2 mL	s.c.	EMEA/V/C/000051

[a] $CCID_{50}$, cell culture infectious dose 50% endpoint; [b] $\log_2$, logarithm in base 2; [c] $TCID_{50}$, tissue culture infectious dose 50% endpoint; [d] VN.U., virus neutralizing antibody titer after vaccination in guinea pigs; i.m., intramuscular route; s.c., subcutaneous route; i.n., intranasal route.

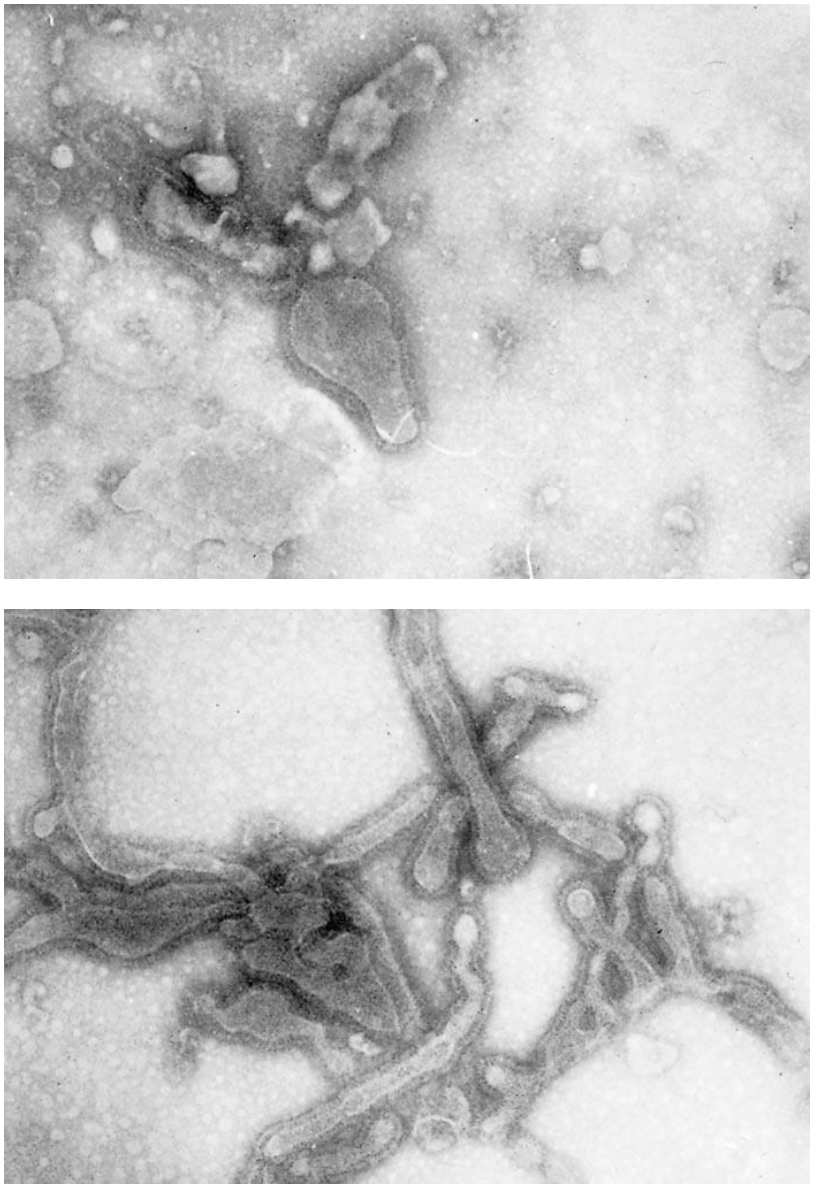
Petrini et al., 2019)

Bovine respiratory syncytial virus (BRSV)

- The cause of a viral pneumonia primarily in calves **under 6 months of age** and also in yearlings and adult cattle.
- Belong to genus *Orthopneumovirus* in the family *Pneumoviridae*
- BRSV is **genetically and antigenically related to human respiratory syncytial virus (HRSV)**.
- Worldwide distribution
  * In Taiwan (1997–1998): 61.9% (468/756) seropositive by ELISA
- **Reinfection is common**; diseases mainly occur **cold seasons**
- One serotype
- Transmission via contact/aerosol



Jean Francois Valarcher & Geraldine Taylor, 2007



https://anipedia.org/resources/bovine-respiratory-syncytial-virus-infection/1018

Pathogenesis

- BRSV replicated in **ciliated respiratory epithelial cells** and **type II pneumocytes**.
- BRSV infection causes airway obstruction and **hyperactivity** that may persist for up to 30 days following viral exposure.
- The virus-specific **IgE** antibody may play a role in the pathogenesis of the severe disease as part of a hypersensitivity reaction.
- RSV can induce extensive pulmonary **mast cell degranulation** and leads to a severe, highly fatal, or ”malignant” form of the disease, **paroxystic respiratory distress syndrome (PRDS).**

Morbidity & Mortality

- In outbreaks, morbidity tends to be high (30% to 50% or higher).
- The mortality rate is usually low but can reach 20%.

https://www.hipra.com/en/detect-and-protect-bovine-respiratory-syncytial-virus

Clinical Presentation

- A sudden outbreak of acute respiratory disease in a group of animals is a characteristic of a primary BRSV infection.
- Older and/or reinfected animals → mild clinical disease
- Clinical signs:
  - ✓Fever (40°–42°C)
  - ✓Depression
  - ✓Decreased feed intake
  - ✓Decreased milk production
  - ✓Increased respiratory rate
  - ✓Cough
  - ✓Nasal and lacrimal discharge
  - →→ Usually resolved within 1-2 weeks

-- 2nd bacterial infection is common
-- A small percentage of cases may develop a fatal interstitial pneumonia.

Gross Findings



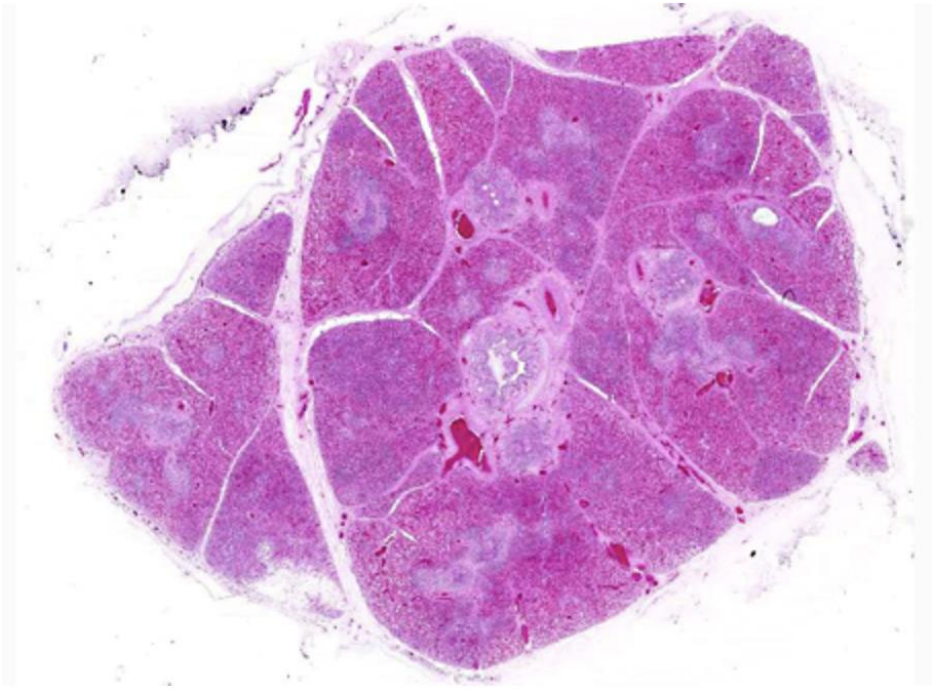
This calf was actuely dyspneic and febrile and demonstrated open mouth breathing with a serous nasal discharge.



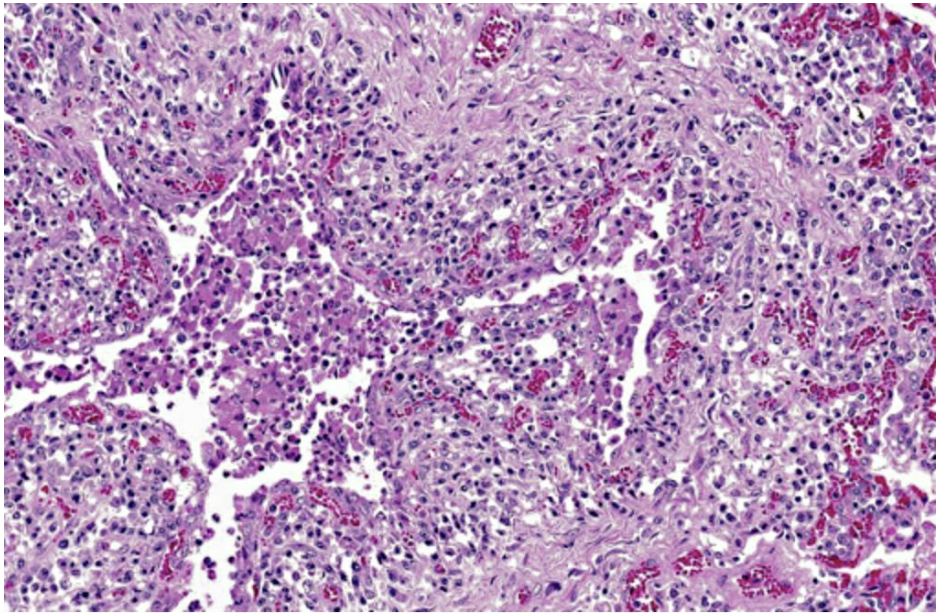
The cranioventral lungs have multifocal to coalescing depressed areas of red-brown consolidation with edema, which were rubbery and firm.

https://www.askjpc.org/wsco/do_slideshow2.php?pic_series=MWFveE1YOWZCeXpwOWdsb3NmVEF6UT09&cYear=2014&initImage=MUJ6QTJDekNqTCtMcGZtT29jWDE0Zz09

Microscopic Features



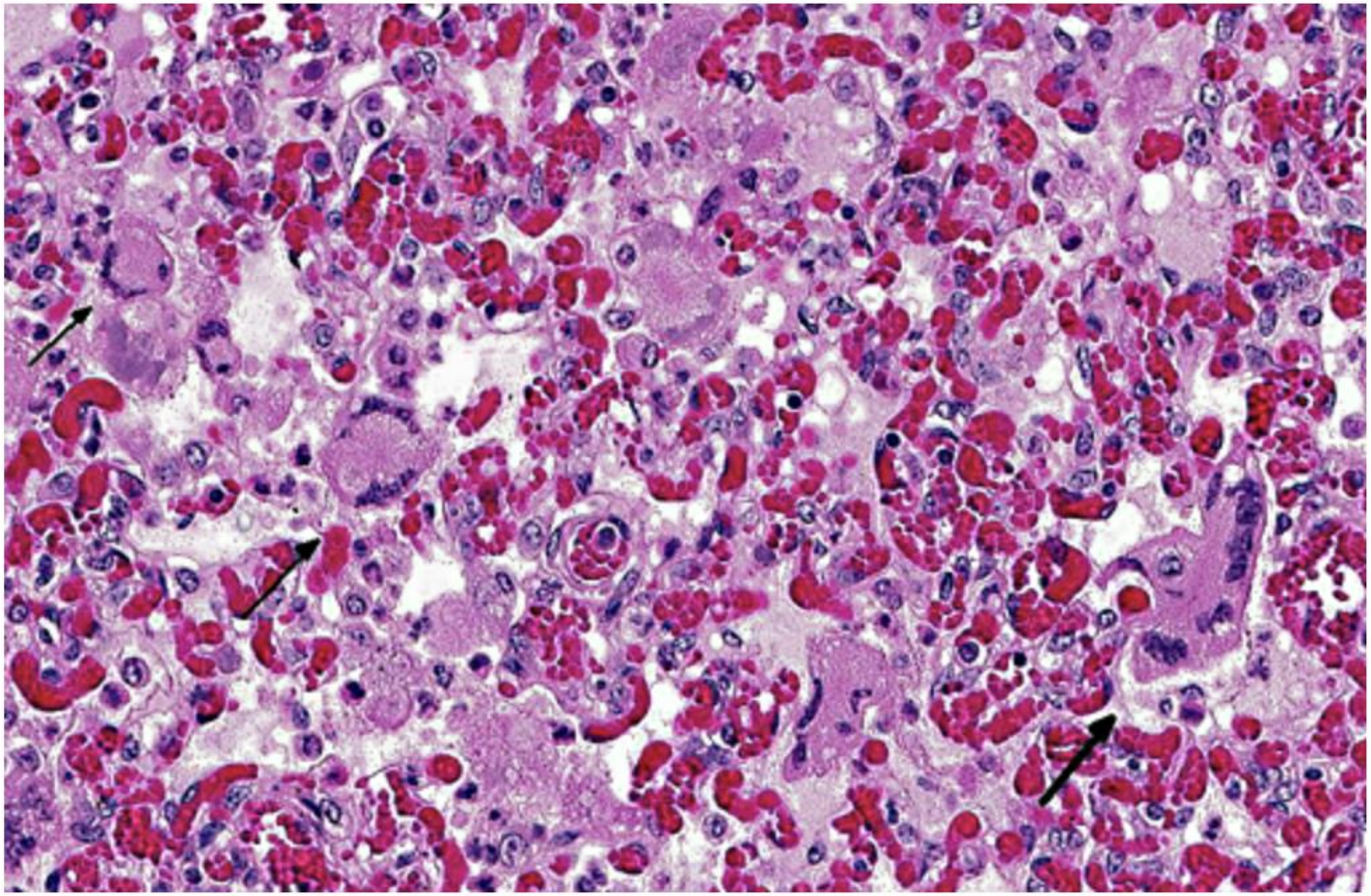
At subgross examination, airways are pale and architecture is distorted (necrosis). The parenchyma is markedly congested and interlobular septa and pleural connective tissue is expanded by edema and emphysema



Bronchiolar epithelium is necrotic or attenuated, and the lumen contains numerous neutrophils and macrophages admixed with cellular debris. Numerous histiocytes and fewer neutrophils, lymphocytes, and plasma cells expand the submucosa.

Bronchiolitis obliterans?

https://www.askjpc.org/wsco/do_slideshow2.php?pic_series=MWFveE1YOWZCeXpwOWdsb3NmVEF6UT09&cYear=2014&initImage=MUJ6QTJDekNqTCtMcGZtT29jWDE0Zz09



1–5. Lung

Hematoxylin and Eosin | 224x

Multifocally, alveolar spaces contain large multinucleated viral syncytia characteristic of bovine respiratory syncytial virus (BRSV).

https://www.askjpc.org/wsco/do_slideshow2.php?pic_series=MWFveE1YOWZCeXpwOWdsb3NmVEF6UT09&cYear=2014&initImage=MUJ6QTJDekNqTCtMcGZtT29jWDE0Zz09

Prevention & Control

- Difficult to control
- Antimicrobial therapy to prevent 2nd bacterial infection.
- Minimize stressors on cattle.
- Vaccinate cattle prior to disease risk; both MLV or killed vaccines are available as part of the multivalent product
- However, efficacy is **controversial**, and given the significant disease burden associated with bRSV infection, there is a definite need for improved technologies.
- Segregate new stock for a few weeks from existing herd and ensure vaccinated against BRSV before mixing.
- Ensure calves get sufficient colostrum from vaccinated dams.

Bovine Parainfluenza virus–3 (BPIV-3)

- Worldwide distribution
- Mainly induces a subclinical to mild, self-limiting disease
- The clinical significance of BPIV-3 is as a trigger leading to the development of secondary bacterial pneumonia.
- Belong to genus *Respirovirus* in the family *Othoparamyxoviridae*
- BPIV-3 is **genetically and antigenically related to human parainfluenza virus type 3.**
- In addition to cattle, BPIV-3 also infect sheep, goats, and wild ruminants, as well as human and non-human primates.
- Transmission via virus-containing secretion/aerosol

Pathogenesis

- BPIV-3 infect ciliated respiratory epithelial cells, pneumocytes and **alveolar macrophages** (PAM).

→Interfering with normal pulmonary clearance mechanisms and allowing secondary invasion by bacteria or mycoplasmas

→decreasing the phagocytic ability of PAM, enhancing the proinflammatory response, and enhancing contraction of respiratory smooth muscle, resulting in bronchoconstriction

Clinical Presentation

- Similar to often milder than those observed in BRSV infection
- Clinical signs:
  - ✓Fever
  - ✓Nasal and lacrimal discharge
  - ✓Depression
  - ✓Increased respiratory rate
  - ✓Cough
  - →→ Usually resolved within 1 weeks

Pathological Findings

- Similar to those observed in BRSV infection, with usually fewer syncytial cells.

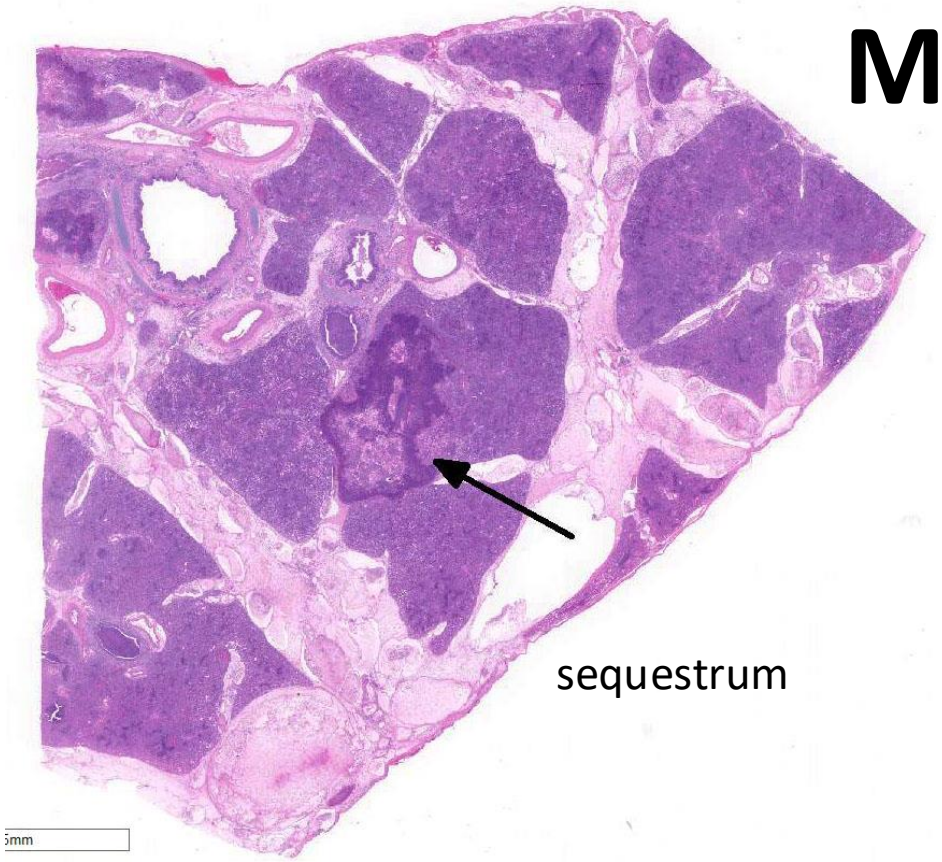
Mannheimia haemolytica

- **Gram-negative** coccobacillus of the *Pasteurellaceae* family
- A **commensal bacterium** in the nasopharynx and tonsillar crypts of immunocompetent ruminants.
- Twelve serotypes (capsular polysaccharide antigens); in a naïve or stressed animal, **serotype 1** is often isolated from pneumonic lungs.
- Stress, cold weather, and co-infection are risk factors of overgrowth and subsequent inhalation of aerosol droplets into the trachea and lungs.

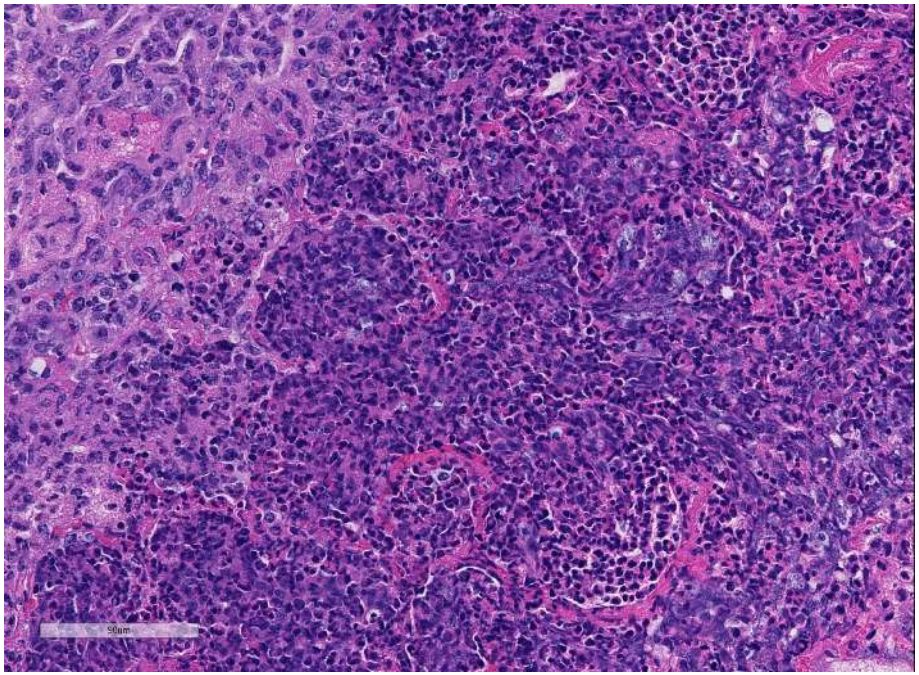
Mannheimia haemolytica

- Many virulence factors including **leukotoxin**, lipopolysaccharide (LPS), adhesins, capsule, outer membrane proteins, and various proteases…
  - ✓**Leukotoxin,** a pore-forming cytolysin is a member of the RTX (repeats in ToXin) family, **recruit** and **activate** leukocytes at low conc. and **lyse** leukocytes and platelets at high conc.
  - ✓The recruited neutrophils are ineffective in killing but damages endothelial cells → **infarct**
  - ✓Lytic and necrotic leukocytes exhibit a streaming pattern of basophilic chromatin → **oat cells**
- Common cause of **fibrinonecrotizing pneumonia** and pleuropneumonia in cattle, sheep, and goats.

Microscopic Features



Subgross examination demonstrates diffuse filling of airways and alveoli with a cellular exudate and a focal area of infarction outlined by a dense band of cellular debris (arrow). The interlobular septa are expanded by edema and fibrin thrombi within lymphatics.



Alveoli are filled with degenerate neutrophils with streaming nuclei (oat cells) admixed with abundant cellular debris.

https://www.askjpc.org/wsco/do_slideshow2.php?pic_series=K25lek15VnBNclcvSjJackdMTTl1Zz09&cYear=2016&initImage=cC9acFdXQjhOWWJFVUl3SGNETFZKQT09

Pasteurella multocida

- **Gram-negative** coccobacillus of the *Pasteurellaceae* family
- Normal inhabitants of the upper- and opportunistic pathogens of the lower respiratory tract of cattle.
- 5 capsular sero- groups (A, B, D, E, and F) and 16 somatic serotypes (1-16)
- Serotype A3 and, to a lesser extent, D3 are the major pathogens associated with lower respiratory tract infection in cattle.
- *P. multocida* produce similar lung lesions in affected cattle with less vascular damage and absence of oat cells.

- *P. multocida* is commonly linked to enzootic/endemic calf pneumonia.



5.12. Endemic calf pneumonia showing mucopurulent nasal discharge



5.13. "Sweaty coat" in endemic calf pneumonia

Control & Prevention BRD

- Reduce stressors that contribute to the development of disease:
  - ✓Minimize transportation time
  - ✓**Precondition** weaned calves before transportation
  - ✓Receive **arrival processing**, ex. vaccination and metaphylatics (eg. long-acting antimicrobials)
  - ✓Ideally, vaccination against pathogens involved in BRD should be given 2-3 w **prior to shipment** and can be repeated on entry to the feedlot.

Other Diseases

Bovine Ephemeral Fever

Bovine Ephemeral Fever (BEF)

mosquitoes and Culicoides biting midges庫蠓(糠蚊)

- An **insect-transmitted**, noncontagious viral disease of cattle
- Caused by BEF virus belonging to the genus *Ephemerovirus* in the family *Rhabdoviridae*
- Enzootic in the tropical areas (including Taiwan and in much of Australia, with extensions into the subtropics and some temperate regions.
- BEF is most prevalent in the **wet and hot seasons** and disappears abruptly in winter.
- **Wind** and **animal transportation** help spread the disease
- High morbidity but low mortality.

Pathogenesis

- BEFV has been isolated from **neutrophils**, **synovial cells** and **reticuloendothelial cells**.

  →generalized inflammation with increased endothelial permeability, vasculitis and thrombosis in serous and synovial cavities.

- Cattle between **6 months and 2 years** of age are more commonly affected.
- Bulls, heavy cattle, and high-lactating dairy cows are the most severely affected

Clinical Presentation

- Sudden onset of **biphasic** to polyphasic fever (40°–42°C)
- Shivering
- Inappetence
- **Serous nasal and ocular discharge**
- Increased heart rate
- Tachypnea or dyspnea
- Muscular signs: listlessness, **stiffness and lameness**
- Sudden **decrease in milk yield**

considerable loss occurs in **dairy herds**

→ Recover within 3 days → also known as **3-day sickness**



calf with high fever unable to stand howdoctor treated /ephemeral fever/3daysfever

D DR YEDLA ANIMAL DOCTOR 訂閱
20.9萬位訂閱者

801 分享 下載 ⋯

https://www.youtube.com/watch?v=2Heqk3HZ2ik&t=19s



圖 1. 感染牛流行熱患牛，臨床上呈發高熱大量流涎症狀 (箭頭)

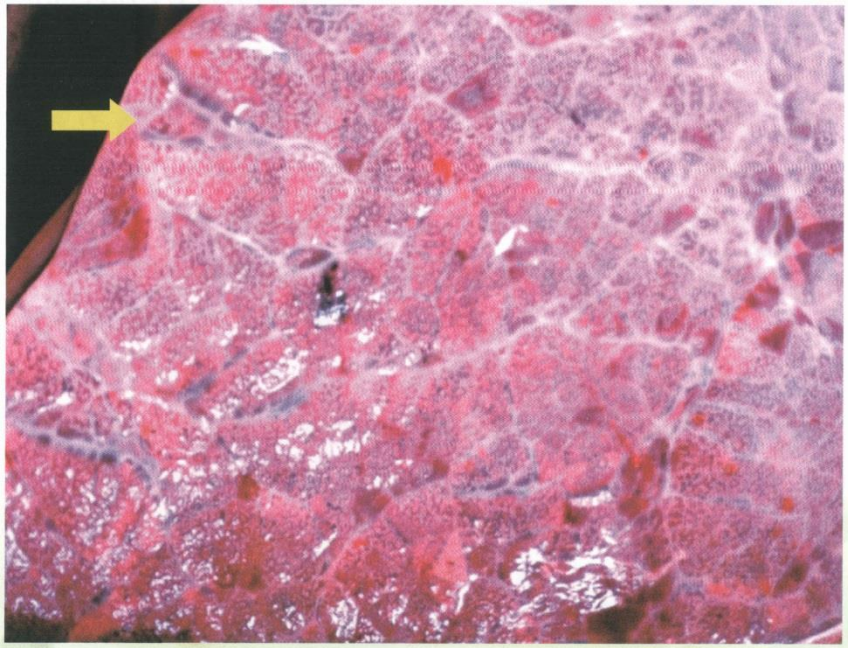


圖 3. 肺氣腫病變近照 (箭頭)

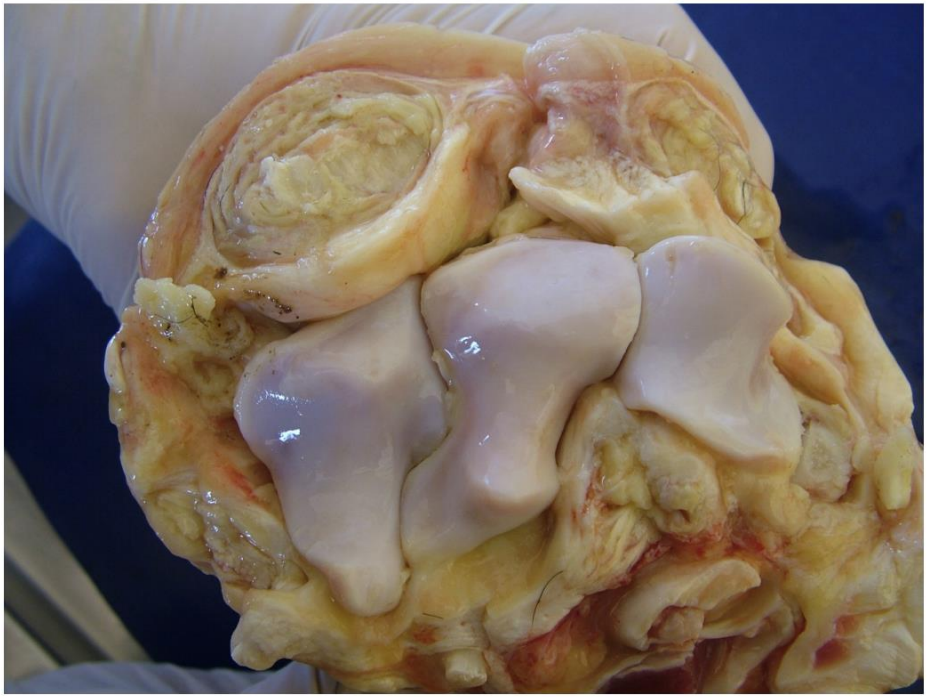


Figure 1 – joint of calf with severe yellowish periarticular gelatinous fluid.

https://www.cpdsolutions.co.za/Publications/article_uploads/Bovine_ephemeral_fever.pdf

Prevention & Control of BEF



- Movement restriction and environmental hygiene
- Vaccination by live attenuated or **inactivated** vaccines are both able to prevent disease in natural outbreaks for periods up to 12 months



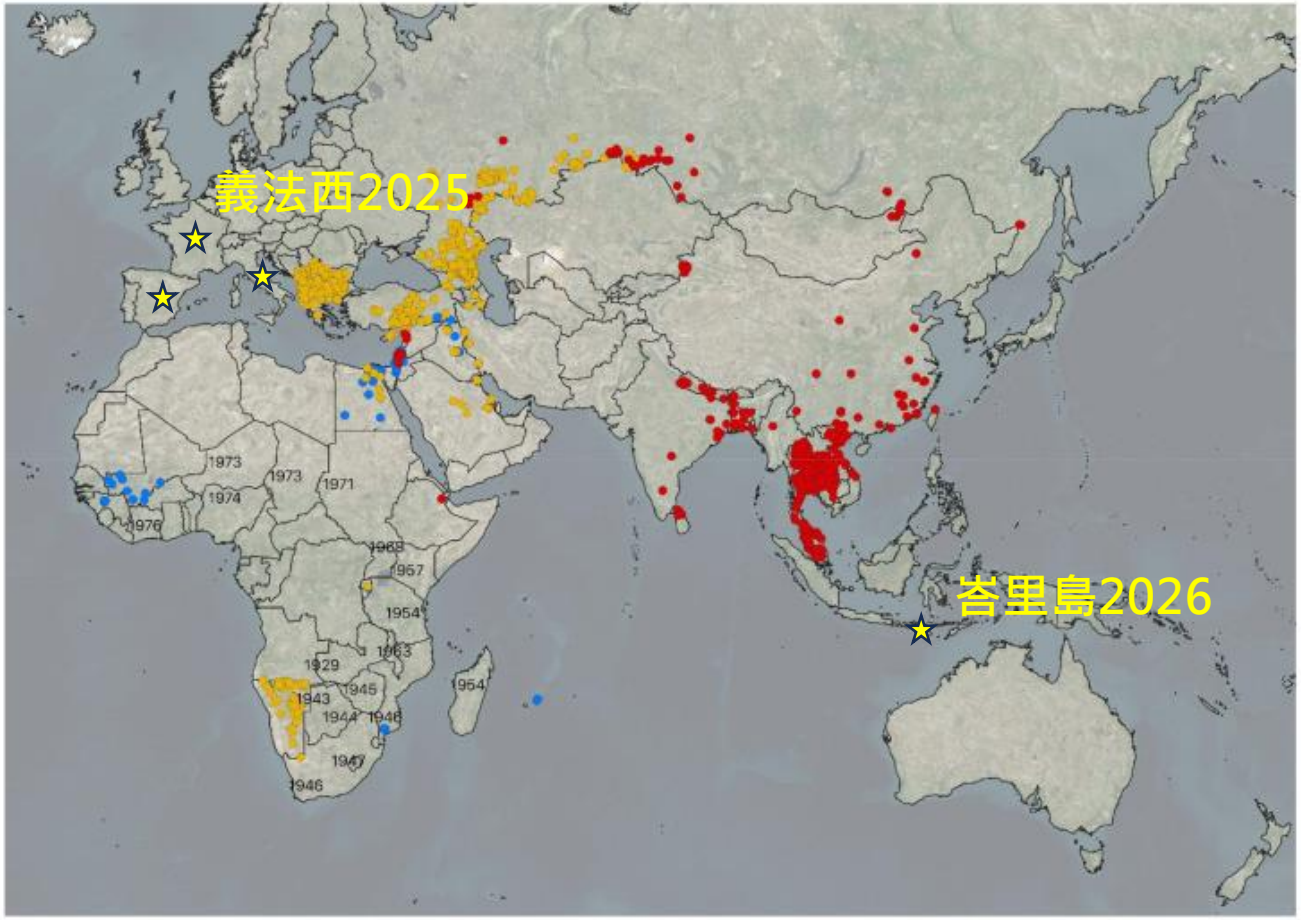
https://www.chinatimes.com/realtimenews/20170616003360-260405?chdtv

Lumpy Skin Disease

Lumpy Skin Disease

- Caused by Lumpy skin disease virus (also known as Neethling poxvirus), of the genus *Capripoxvirus*.
- WOAH notifiable disease
- Significant economic impact due to **reduced milk** production, reduced body condition, hide damage, decreased fertility in bulls, and abortion in cows.
- Transmission by a range of sucking/biting **arthropod vectors** and potentially by contact.
- Variably morbidity (5%–50%); usually low mortality.
- Highest incidence during **wet summer** weather and outbreaks tend to follow **waterways**.

Epidemiology



- Endemic in most African countries.
- Spread to the Middle East and Asian countries since 2012.

Outbreaks marked with blue dots were reported in 2006–2013, yellow in 2014–2018, and red in 2019–2021.

Tuppurainen et al., 2023

Clinical Presentation

- The incubation period is typically 2 to 4 weeks in field outbreaks.
- Clinical signs (vary from inapparent to severe):
  - ✓Fever
  - ✓Nasal discharge, lacrimation and hypersalivation
  - ✓Characteristic **eruptions of skin nodules**, often initially on the perineum (會陰)
- **Recovery usually takes 4 to 12 weeks, and pregnant cows may abort**.



Clinical signs of lumpy skin disease in cattle in northern provinces of Vietnam. A and B: Widespread, firm skin nodules in an affected cattle and calf; B and C: Conjunctivitis, depression, anorexia in a calf; D: Affected cattle with sitfast lesions characterized by large necrotic skin nodules (black arrows).

Bich et al., 2023

Prevention & Control

- Control of cattle movement
- Freeze-dried, **live attenuated vaccines** (either based on LSDV, sheep poxvirus or goat pox virus) are commercially available and the most commonly used.
- Slaughter of affected and in-contact animals, destruction of contaminated hides, and vaccination of at-risk animals is a common approach when the disease is introduced to a previously free country.
- Successful control and eradication of LSD relies on **early detection** of the index case, followed by a rapid and widespread **vaccination** campaign.



https://www.ey.gov.tw/Page/9277F759E41CCD91/14337d30-0406-4909-8dc3-348b973ccef5

2021.4中旬台灣新北市林口肉牛場 LSD

- 發病場撲殺病牛，移動管制；病毒與金門 LSD 同源；
- 周邊半徑3公里內養牛場移動管制，加強訪視；
- 全島施打減毒疫苗，牛場訪視。
- 減毒疫苗施打後約 1/3牛出現**全身性皮膚結節，但 size 小於發病牛，乳牛泌乳量下降約一週。**





農科院陳世平 博士提供

Slide provided by 鄭益謙老師

Rinderpest (cattle plaque)

- Caused by rinderpest virus belonging to the genus ***morbillivirus***
- Declared **globally eradicated** in 2011 by OIE/WOAH.
- Transmission required direct or close indirect contact
- Often occurred as epizootics associated with a **very high morbidity & mortality rate** due to severe **diarrhea/dysentery** and dehydration.

Think about CD

- The virus infect mucosal epithelial cells, monocyte and lymphocytes → mucosal erosion/ulceration & lymphoid necrosis
- Clinical signs include fever, anorexia, depression, and oculonasal discharges developed, followed by necrotic lesions on the gums, buccal mucosa, and tongue.
- Affected animals develop immunosuppression and are prone to 2nd bacterial infection
- Recovery may take weeks.

Bovine Tuberculosis

Bovine Tuberculosis (Bovine TB)

- Primarily by **M. bovis**, but also by M. caprae and to a lesser extent M. tuberculosis.
  -- Member of the Mycobacterium tuberculosis complex (MTBC)
  -- Acid-fast, slow-growing intracellular bacterium
- Chronic granulomatous disease
- Broad host range
  -- Cattle are the major domestic reservoir
  -- Wildlife reservoirs are important in some countries
- **Zoonotic**
- **WOAH-listed disease**

Transmission

- Among cattle:
  -- Primarily respiratory route
  -- Less commonly ingestion
- In other species:
  -- Aerosol/direct contact
  -- consumption of contaminated unpasteurized milk



https://rr-asia.woah.org/en/projects/bovine-tuberculosis/

雲嘉傳53頭乳牛染結核病 鮮乳安全現疑慮

#結核病 #檢測 #農委會 #陽性 ...



讚 0

發布時間：2017/4/20 19:41 更新時間：2017/4/20 20:51

嘉義牛場傳出乳牛驗出牛結核病，農委會防檢局傍晚證實這個消息！雲林防疫所指出，口湖鄉有業者在一月下旬想結束營業，把兩百多隻乳牛賣到嘉義，三月分陸續檢測，確診其中53頭檢出陽性。至於檢測陰性的牛隻，仍然可以正常產乳，也強調目前市售的鮮乳安全沒有問題，可以放心食用。 照片中，防疫人員在牛場忙著消毒，就怕疫情持續擴大！農委會防檢局



Cats & dogs

TB incidence is low in cats and very low in dogs. Because bovine TB is a zoonotic disease (i.e. it can be spread from animal to human), where TB is found in pets, APHA or private vets will inform the Local Health Authority so that any risks to human contacts can be investigated.

If your pet is infected with TB it may show signs of disease which include coughing, wheezing and/or weight loss. Lumps, abscesses or bite wounds which fail to heal, especially those around the head and neck, can also be caused by TB and are most frequently seen in infected cats.

M.bovis infection in cats is predominantly seen as a cutaneous (skin) form. The sites affected are frequently the sites of bite wounds such as the limbs, face and tail head, and the lesions may take the form of raised nodules or non-healing ulcers. The most commonly affected lymph node is the submandibular node. Respiratory and alimentary (gut) forms are seen less frequently. The alimentary form may be seen when cats are fed raw milk from TB infected cows.

M.bovis infection is much rarer in dogs than cats. It may be associated with bite wounds from infected wildlife. Non-healing bite wounds and enlarged calcified (gritty) lymph nodes have been described. Granulomatous lesions have also been described in the lungs, pleura, liver, kidneys and lymph nodes.

Treatment of TB infected pets is not recommended because of the risk this presents of transmitting the disease to other animals and/or the pet's owner. If TB is reported in a farm cat or dog, APHA will instigate TB testing of any cattle on the farm and other, potentially exposed cattle, on neighbouring premises.

Testing tissue samples:

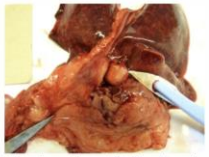
There is a legal requirement to report all suspicions of TB infection in the carcases of farmed animals and pets to APHA. Confirmation of

Cookie Settings

Post-mortem images of TB in cats and dogs



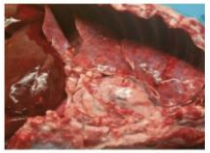
A non-healing ulcerated lesion on the foot of a cat (image source: APHA)



An enlarged bronchial lymph node in a cat with TB (image source: APHA)



Enlarged caudal mesenteric lymph node in a cat with TB. The cat was a farm cat which had been fed raw milk (image source: APHA)



Granulomatous lesions affecting the lungs and pleura of a dog with TB (image source: APHA)

Clinical & Pathological Findings

Clinical Presentation

- Usually **chronic and progressive**
  -- Many infected cattle remain asymptomatic
  -- Advanced disease:
  Progressive weight loss / emaciation
  Weakness
  Chronic cough
  Enlarged lymph nodes
  Reduced production

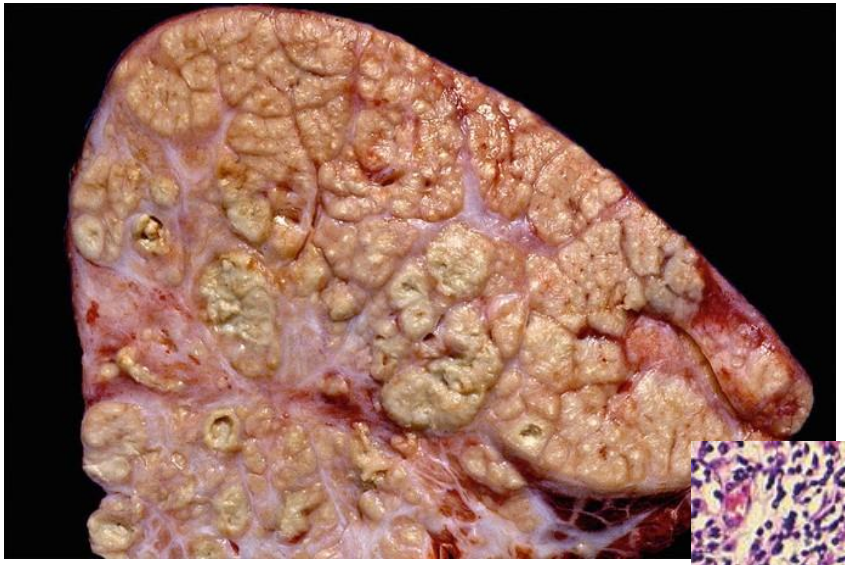
Characteristic lesion:

Think about the transmission route!

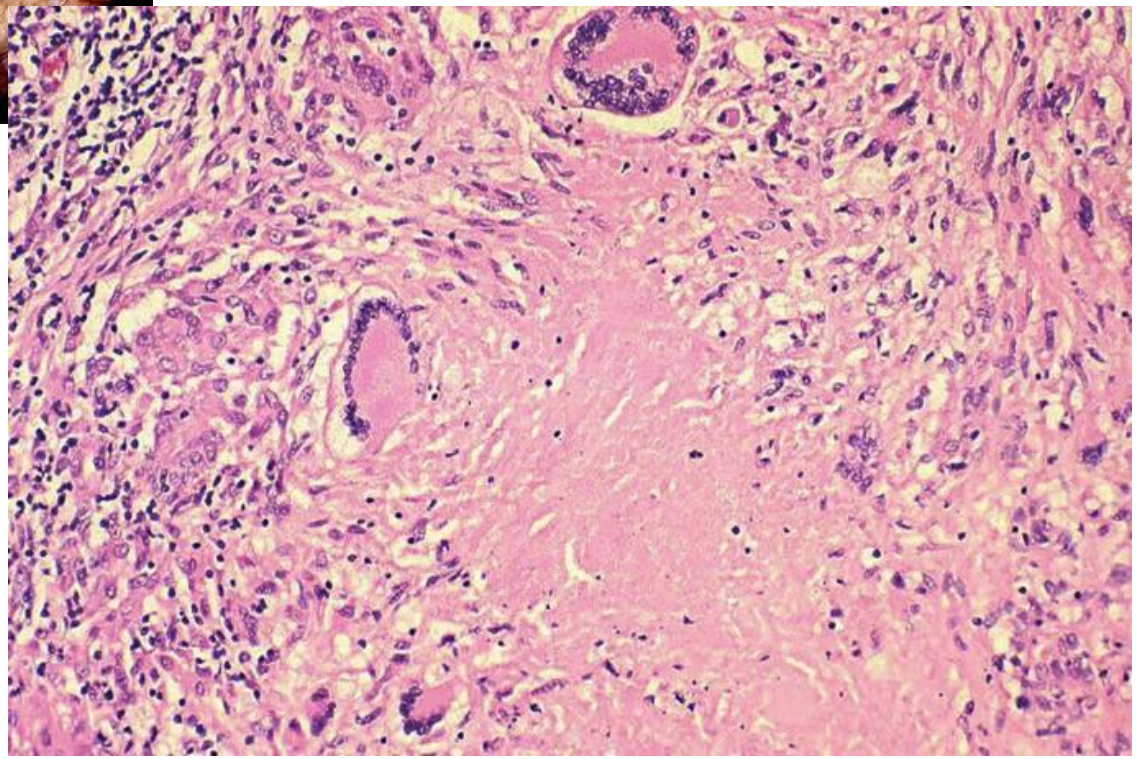
- **Granuloma (tubercle)**
  -- Common sites: **Retropharyngeal/bronchial/mediastinal LN**
  -- Lung: Firm, pale-yellow/white nodules
  -- **Caseous necrosis ± mineralization**



消瘦的牛



Multifocal to coalescing granulomas



Histopathological characteristics of a typical tuberculous granuloma. Note the central caseous necrosis and the presence of a number of Langhans giant cells, epithelioid cells, loosely arranged lymphocytes, and early fibrosis at the periphery of the lesion

Diagnosis & Control

Ante-mortem diagnosis

- **Tuberculin skin test**
  -- Measures delayed-type hypersensitivity (Type IV)
  -- Intradermal injection of PPD
  -- Read at approximately 72 h

➢Single intradermal tuberculin test (SITT)
  -- bovine PPD

➢Comparative intradermal tuberculin test (CITT)
  -- bovine PPD + avian PPD → comparison helps distinguish sensitization to MTBC from environmental/nontuberculous mycobacteria

- IFN-γ release assay
  -- Measures cell-mediated immune response in blood
  -- Can complement skin testing

Diagnosis & Control

Post-mortem diagnosis

- Histopathology / acid-fast stain
- PCR
- Mycobacterial culture → definitive identification